

Sunscreen Survey Analysis - Data Cleaning and Preprocessing + Part of EDA

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2026-02-02

Libraries

```
library(janitor)
```

```
##  
## Attaching package: 'janitor'  
  
## The following objects are masked from 'package:stats':  
##  
##   chisq.test, fisher.test
```

```
library(ggplot2)  
library(vcd)
```

```
## Loading required package: grid
```

```
library(gmodels)  
library(reshape2)  
library(ggpubr)  
library(forcats)  
library(MASS)  
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v stringr  1.5.1  
## v lubridate 1.9.4      v tibble   3.3.0  
## v purrr      1.1.0      v tidyr    1.3.1  
## v readr      2.1.5
```

```
## -- Conflicts ----- tidyverse_conflicts() --  
## x dplyr::filter() masks stats::filter()  
## x dplyr::lag()     masks stats::lag()  
## x dplyr::select() masks MASS::select()  
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(car)
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
##
##     some
```

1. Data Cleaning, Transformation, and Reduction

Read in Dataset

```
data <- read.csv("data/SunscreenSurveySBU_DATA_2025-06-18_2022 - SunscreenSurveySBU_DATA_2025-06-18_2022.csv")
glimpse(data)
```

```
## Rows: 1,008
## Columns: 67
## $ participant_id      <int> 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, ~
## $ redcap_survey_identifier <lg1> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ consent_timestamp    <chr> "2025-03-24 10:16:02", "2025-04-02 8:35:41"~
## $ key_information_the_inform <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1~
## $ sign_consent_below   <chr> "signature_2025-03-24_1015.png", "signature~
## $ consent_complete     <int> 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2~
## $ survey_timestamp     <chr> "2025-03-24 10:18:11", "2025-04-02 8:37:44"~
## $ school               <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, NA, ~
## $ other_school         <chr> "", "", "", "", "", "", "", "", "", "", "", ~
## $ gender               <int> 2, 2, 2, 2, 2, 2, 2, 2, 1, 2, 2, 2, 1, NA, ~
## $ gender_other         <chr> "", "", "", "", "", "", "", "", "", "", "", ~
## $ seen_free_dispenser  <int> 1, 2, 2, 2, 1, 2, 2, 1, 1, 2, 2, 2, 1, NA, ~
## $ would_you_use        <int> 3, 3, 1, 3, 2, 2, 1, 1, 1, 1, 1, 2, 1, NA, ~
## $ aid_in_prevention    <int> 1, 1, 1, 3, 3, 1, 3, 1, 1, 1, 1, 1, 1, NA, ~
## $ where___1           <int> 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0~
## $ where___2           <int> 1, 1, 1, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1, 0~
## $ where___3           <int> 0, 1, 0, 0, 1, 0, 1, 1, 0, 1, 1, 1, 0, 0~
## $ where___4           <int> 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0~
## $ where___5           <int> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0~
## $ where___6           <int> 1, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0~
## $ where___7           <int> 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0~
## $ where___8           <int> 0, 1, 1, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0, 0~
## $ where___9           <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ other_location       <chr> "", "", "", "", "", "", "", "", "", "", "", ~
## $ skin_type            <int> 3, 2, 3, 3, 4, 1, 4, 4, 3, 2, 1, 2, 2, NA, ~
## $ apply_freq           <int> 2, 2, 2, 4, 2, 2, 4, 3, 2, 2, 2, 1, 1, NA, ~
## $ winter___1          <int> 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 1~
## $ winter___2          <int> 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ winter___3          <int> 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0~
```

```
## $ winter___4 <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ spring___1 <int> 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1~
## $ spring___2 <int> 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 1, 0, 0~
## $ spring___3 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0~
## $ spring___4 <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ summer___1 <int> 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0, 1, 0, 1~
## $ summer___2 <int> 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0~
## $ summer___3 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ summer___4 <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0~
## $ fall___1 <int> 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0, 1, 0, 1~
## $ fall___2 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ fall___3 <int> 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0~
## $ fall___4 <int> 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0~
## $ price <int> 2, 1, 2, 1, 2, 2, 2, 1, 1, 1, 2, 1, 1, NA, ~
## $ body_part___1 <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1~
## $ body_part___2 <int> 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1~
## $ body_part___3 <int> 1, 0, 1, 1, 1, 0, 1, 1, 0, 0, 1, 1, 0, 0, 0~
## $ body_part___4 <int> 1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0~
## $ body_part___5 <int> 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0~
## $ body_part___6 <int> 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0~
## $ body_part___7 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0~
## $ body_part___8 <int> 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0~
## $ body_part___9 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0~
## $ body_part___10 <int> 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 0~
## $ body_part___11 <int> 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0~
## $ body_part___12 <int> 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 0, 0~
## $ body_part___13 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0~
## $ incentive <int> 3, 1, 3, 3, 4, 3, 2, 3, 1, 1, 3, 2, 1, NA, ~
## $ other_incentive <chr> "", "", "", "", "Prevent acne scarring/dark~
## $ reasons_against___1 <int> 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 0~
## $ reasons_against___2 <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ reasons_against___3 <int> 0, 1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0~
## $ reasons_against___4 <int> 1, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 1~
## $ reasons_against___5 <int> 1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ reasons_against___6 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ other_reason_against <chr> "", "", "", "", "", "", "", "", "", "", "", ~
## $ diversity <int> 2, 2, 2, 2, 3, 2, 2, 2, 2, 1, 2, 2, 2, NA, ~
## $ survey_complete <int> 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 0, 2~
```

Remove unwanted columns and combine sparse school levels

```
data_trim <- data %>%
  janitor::clean_names() %>%
  # remove any column whose name contains "other"
  dplyr::select(-matches("other", ignore.case = TRUE)) %>%
  # drop first 7 columns and the last column
  dplyr::select(-(1:7), -last_col()) %>%
  # collapse healthcare school into 1 group (numeric only, no labels yet)
  dplyr::mutate(
    school = dplyr::case_when(
      .data$school == 1L ~ 1L, # Undergraduate
      .data$school == 2L ~ 2L, # Graduate
      .data$school %in% c(3L,4L,5L,6L, 7L) ~ 3L, # Healthcare students collapsed with Other
      TRUE ~ NA_integer_
```

```

    )
  )
rm(data) # drop the original dataset to avoid conflicts

glimpse(data_trim)

```

```

## Rows: 1,008
## Columns: 54
## $ school      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, NA, 3, 1, 3~
## $ gender      <int> 2, 2, 2, 2, 2, 2, 2, 2, 1, 2, 2, 2, 1, NA, 2, 2, N~
## $ seen_free_dispenser <int> 1, 2, 2, 2, 1, 2, 2, 1, 1, 2, 2, 2, 1, NA, 2, 1, 2~
## $ would_you_use <int> 3, 3, 1, 3, 2, 2, 1, 1, 1, 1, 1, 2, 1, NA, 1, 1, 1~
## $ aid_in_prevention <int> 1, 1, 1, 3, 3, 1, 3, 1, 1, 1, 1, 1, 1, NA, 1, 1, 1~
## $ where_1      <int> 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 1, 0,~
## $ where_2      <int> 1, 1, 1, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 0,~
## $ where_3      <int> 0, 1, 0, 0, 1, 0, 1, 1, 0, 1, 1, 1, 0, 0, 0, 0, 0,~
## $ where_4      <int> 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ where_5      <int> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0,~
## $ where_6      <int> 1, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0,~
## $ where_7      <int> 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 1,~
## $ where_8      <int> 0, 1, 1, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0, 0, 0, 1, 0,~
## $ where_9      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ skin_type    <int> 3, 2, 3, 3, 4, 1, 4, 4, 3, 2, 1, 2, 2, NA, 3, 5, 1~
## $ apply_freq   <int> 2, 2, 2, 4, 2, 2, 4, 3, 2, 2, 2, 1, 1, NA, 2, 2, N~
## $ winter_1     <int> 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 0,~
## $ winter_2     <int> 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1,~
## $ winter_3     <int> 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0,~
## $ winter_4     <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ spring_1     <int> 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 0,~
## $ spring_2     <int> 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 1, 0, 0, 1, 1,~
## $ spring_3     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ spring_4     <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ summer_1     <int> 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0, 1, 0, 1, 1, 1,~
## $ summer_2     <int> 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0,~
## $ summer_3     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ summer_4     <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ fall_1       <int> 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0, 1, 0, 1, 0, 0,~
## $ fall_2       <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1,~
## $ fall_3       <int> 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0,~
## $ fall_4       <int> 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ price        <int> 2, 1, 2, 1, 2, 2, 2, 1, 1, 1, 2, 1, 1, NA, 1, 2, 1~
## $ body_part_1  <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1,~
## $ body_part_2  <int> 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 0, 0,~
## $ body_part_3  <int> 1, 0, 1, 1, 1, 0, 1, 1, 0, 0, 1, 1, 0, 0, 0, 0, 0,~
## $ body_part_4  <int> 1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ body_part_5  <int> 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ body_part_6  <int> 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0,~
## $ body_part_7  <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ body_part_8  <int> 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ body_part_9  <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ body_part_10 <int> 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0,~
## $ body_part_11 <int> 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0,~
## $ body_part_12 <int> 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0,~

```

```
## $ body_part_13      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0,~
## $ incentive         <int> 3, 1, 3, 3, 4, 3, 2, 3, 1, 1, 3, 2, 1, NA, 2, 2, 1,~
## $ reasons_against_1 <int> 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0,~
## $ reasons_against_2 <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0,~
## $ reasons_against_3 <int> 0, 1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0,~
## $ reasons_against_4 <int> 1, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1,~
## $ reasons_against_5 <int> 1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ reasons_against_6 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ diversity         <int> 2, 2, 2, 2, 3, 2, 2, 2, 2, 2, 1, 2, 2, 2, NA, 2, 2, 2,~
```

Recode binary and Yes, No, Maybe ordinal variables

```
recode_ynm_ord <- function(x){
  dplyr::recode(x,
    `1` = 3L, # Yes = 3
    `2` = 1L, # No  = 1
    `3` = 2L, # Maybe = 2
    .default = NA_integer_)
}

recode_yesno_bin <- function(x){
  dplyr::recode(x,
    `1` = 1L, # Yes = 1
    `2` = 0L, # No  = 0
    .default = NA_integer_)
}
```

```
data_clean <- data_trim %>%
  mutate(
    # recode
    would_you_use = recode_ynm_ord(would_you_use),
    aid_in_prevention = recode_ynm_ord(aid_in_prevention),
    seen_free_dispenser = recode_yesno_bin(seen_free_dispenser),
    price = recode_yesno_bin(price)
  )
```

```
glimpse(data_clean)
```

```
## Rows: 1,008
## Columns: 54
## $ school      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, NA, 3, 1, 3~
## $ gender      <int> 2, 2, 2, 2, 2, 2, 2, 2, 1, 2, 2, 2, 1, NA, 2, 2, N~
## $ seen_free_dispenser <int> 1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 0, 0, 1, NA, 0, 1, 0~
## $ would_you_use <int> 2, 2, 3, 2, 1, 1, 3, 3, 3, 3, 3, 3, 1, 3, NA, 3, 3, 3~
## $ aid_in_prevention <int> 3, 3, 3, 2, 2, 3, 2, 3, 3, 3, 3, 3, 3, NA, 3, 3, 3~
## $ where_1      <int> 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 1, 0,~
## $ where_2      <int> 1, 1, 1, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 0,~
## $ where_3      <int> 0, 1, 0, 0, 1, 0, 1, 1, 0, 1, 1, 1, 0, 0, 0, 0, 0,~
## $ where_4      <int> 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ where_5      <int> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0,~
## $ where_6      <int> 1, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0,~
## $ where_7      <int> 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 1,~
## $ where_8      <int> 0, 1, 1, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0, 0, 0, 1, 0,~
```

```
## $ where_9      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ skin_type    <int> 3, 2, 3, 3, 4, 1, 4, 4, 3, 2, 1, 2, 2, NA, 3, 5, 1, ~
## $ apply_freq   <int> 2, 2, 2, 4, 2, 2, 4, 3, 2, 2, 2, 1, 1, NA, 2, 2, N~
## $ winter_1     <int> 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 0, ~
## $ winter_2     <int> 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, ~
## $ winter_3     <int> 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, ~
## $ winter_4     <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ spring_1     <int> 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 0, ~
## $ spring_2     <int> 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 1, 0, 0, 1, 1, ~
## $ spring_3     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, ~
## $ spring_4     <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ summer_1     <int> 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0, 1, 0, 1, 1, 1, ~
## $ summer_2     <int> 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, ~
## $ summer_3     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ summer_4     <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, ~
## $ fall_1       <int> 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0, 1, 0, 1, 0, 0, ~
## $ fall_2       <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, ~
## $ fall_3       <int> 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, ~
## $ fall_4       <int> 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ price        <int> 0, 1, 0, 1, 0, 0, 0, 1, 1, 1, 0, 1, 1, NA, 1, 0, 1, ~
## $ body_part_1  <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, ~
## $ body_part_2  <int> 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 0, 0, ~
## $ body_part_3  <int> 1, 0, 1, 1, 1, 0, 1, 1, 0, 0, 1, 1, 0, 0, 0, 0, 0, ~
## $ body_part_4  <int> 1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ body_part_5  <int> 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, ~
## $ body_part_6  <int> 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, ~
## $ body_part_7  <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, ~
## $ body_part_8  <int> 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, ~
## $ body_part_9  <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, ~
## $ body_part_10 <int> 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, ~
## $ body_part_11 <int> 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, ~
## $ body_part_12 <int> 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, ~
## $ body_part_13 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, ~
## $ incentive    <int> 3, 1, 3, 3, 4, 3, 2, 3, 1, 1, 3, 2, 1, NA, 2, 2, 1, ~
## $ reasons_against_1 <int> 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, ~
## $ reasons_against_2 <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, ~
## $ reasons_against_3 <int> 0, 1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, ~
## $ reasons_against_4 <int> 1, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 1, 1, 1, ~
## $ reasons_against_5 <int> 1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ reasons_against_6 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ diversity    <int> 2, 2, 2, 2, 3, 2, 2, 2, 2, 2, 1, 2, 2, 2, NA, 2, 2, 2~
```

Check if more than one application frequency was selected per season per respondent

```
library(dplyr)

# Seasonal column groups
winter_cols <- c("winter_1", "winter_2", "winter_3", "winter_4")
spring_cols <- c("spring_1", "spring_2", "spring_3", "spring_4")
summer_cols <- c("summer_1", "summer_2", "summer_3", "summer_4")
fall_cols   <- c("fall_1", "fall_2", "fall_3", "fall_4")

# Summarize the sums per respondent for each season
season_check <- data_clean %>%
```

```
mutate(
  winter_sum = rowSums(dplyr::select(., all_of(winter_cols)), na.rm = TRUE),
  spring_sum = rowSums(dplyr::select(., all_of(spring_cols)), na.rm = TRUE),
  summer_sum = rowSums(dplyr::select(., all_of(summer_cols)), na.rm = TRUE),
  fall_sum = rowSums(dplyr::select(., all_of(fall_cols)), na.rm = TRUE)
) %>%
summarise(
  winter_invalid = sum(!winter_sum %in% c(0, 1)),
  spring_invalid = sum(!spring_sum %in% c(0, 1)),
  summer_invalid = sum(!summer_sum %in% c(0, 1)),
  fall_invalid = sum(!fall_sum %in% c(0, 1))
)

print(season_check)
```

```
##   winter_invalid spring_invalid summer_invalid fall_invalid
## 1              8             21             26             15
```

List out occurrences of more than one application frequency type selected per season per respondent

```
data_clean %>%
  mutate(
    winter_sum = rowSums(dplyr::select(., all_of(winter_cols)), na.rm = TRUE),
    spring_sum = rowSums(dplyr::select(., all_of(spring_cols)), na.rm = TRUE),
    summer_sum = rowSums(dplyr::select(., all_of(summer_cols)), na.rm = TRUE),
    fall_sum = rowSums(dplyr::select(., all_of(fall_cols)), na.rm = TRUE)
  ) %>%
  mutate(row_num = row_number()) %>%
  filter(winter_sum > 1 | spring_sum > 1 | summer_sum > 1 | fall_sum > 1) %>%
  dplyr::select(row_num, winter_sum, spring_sum, summer_sum, fall_sum)
```

```
##   row_num winter_sum spring_sum summer_sum fall_sum
## 1      38         2         2         2         1
## 2      52         1         1         2         1
## 3      77         1         2         2         1
## 4      80         1         1         2         2
## 5      81         1         1         2         1
## 6      97         1         2         1         1
## 7     154         1         1         2         1
## 8     168         1         2         2         2
## 9     180         1         1         2         1
## 10    184         1         2         2         1
## 11    218         1         1         2         1
## 12    231         2         1         1         1
## 13    308         1         1         2         2
## 14    317         2         1         2         1
## 15    322         1         2         1         2
## 16    329         1         2         2         2
## 17    358         1         2         1         2
## 18    363         1         2         1         1
## 19    385         1         1         1         2
```

## 20	423	2	1	1	2
## 21	438	1	2	2	1
## 22	496	2	1	1	1
## 23	510	0	1	2	1
## 24	569	2	2	1	1
## 25	634	1	2	1	1
## 26	656	1	1	2	1
## 27	666	1	1	2	1
## 28	672	1	2	2	2
## 29	751	0	2	1	1
## 30	761	1	1	1	2
## 31	783	1	2	1	1
## 32	823	1	1	2	1
## 33	830	1	2	1	1
## 34	856	1	1	2	2
## 35	863	1	1	1	2
## 36	873	1	1	2	2
## 37	878	1	1	2	1
## 38	884	2	2	1	2
## 39	894	1	2	1	1
## 40	910	1	2	1	1
## 41	919	2	2	1	2
## 42	921	1	1	2	1
## 43	922	1	2	1	1
## 44	930	1	1	2	1
## 45	944	0	0	3	0
## 46	955	1	1	2	1

Create merged columns of each season with row values indicating application frequency

```
winter_cols <- c("winter_1", "winter_2", "winter_3", "winter_4")
spring_cols <- c("spring_1", "spring_2", "spring_3", "spring_4")
summer_cols <- c("summer_1", "summer_2", "summer_3", "summer_4")
fall_cols   <- c("fall_1", "fall_2", "fall_3", "fall_4")

collapse_season_numeric <- function(df, cols) {
  idx <- apply(df[, cols], 1, function(x) {
    s <- sum(x == 1, na.rm = TRUE)
    if (s == 1) which(x == 1)[1] else NA_integer_
  })
  # Map index (1..4) to the same survey order
  as.integer(ifelse(is.na(idx), NA_integer_, idx))
}

data_with_seasons <- data_clean %>%
  dplyr::mutate(
    winter_apply_freq = collapse_season_numeric(., winter_cols),
    spring_apply_freq = collapse_season_numeric(., spring_cols),
    summer_apply_freq = collapse_season_numeric(., summer_cols),
    fall_apply_freq   = collapse_season_numeric(., fall_cols)
  )
```

```
glimpse(data_with_seasons)
```

```
## Rows: 1,008
## Columns: 58
## $ school      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, NA, 3, 1, 3~
## $ gender      <int> 2, 2, 2, 2, 2, 2, 2, 2, 1, 2, 2, 2, 1, NA, 2, 2, N~
## $ seen_free_dispenser <int> 1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 0, 0, 1, NA, 0, 1, 0~
## $ would_you_use <int> 2, 2, 3, 2, 1, 1, 3, 3, 3, 3, 3, 1, 3, NA, 3, 3, 3~
## $ aid_in_prevention <int> 3, 3, 3, 2, 2, 3, 2, 3, 3, 3, 3, 3, 3, NA, 3, 3, 3~
## $ where_1      <int> 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 1, 0,~
## $ where_2      <int> 1, 1, 1, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 0,~
## $ where_3      <int> 0, 1, 0, 0, 1, 0, 1, 1, 0, 1, 1, 1, 0, 0, 0, 0, 0,~
## $ where_4      <int> 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ where_5      <int> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0,~
## $ where_6      <int> 1, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0,~
## $ where_7      <int> 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 1,~
## $ where_8      <int> 0, 1, 1, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0, 0, 0, 1, 0,~
## $ where_9      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ skin_type    <int> 3, 2, 3, 3, 4, 1, 4, 4, 3, 2, 1, 2, 2, NA, 3, 5, 1~
## $ apply_freq   <int> 2, 2, 2, 4, 2, 2, 4, 3, 2, 2, 2, 1, 1, NA, 2, 2, N~
## $ winter_1     <int> 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 0,~
## $ winter_2     <int> 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1,~
## $ winter_3     <int> 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0,~
## $ winter_4     <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ spring_1     <int> 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 0,~
## $ spring_2     <int> 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 1, 0, 0, 1, 1,~
## $ spring_3     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ spring_4     <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ summer_1     <int> 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0, 1, 0, 1, 1, 1,~
## $ summer_2     <int> 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0,~
## $ summer_3     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ summer_4     <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ fall_1       <int> 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0, 1, 0, 1, 0, 0,~
## $ fall_2       <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1,~
## $ fall_3       <int> 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0,~
## $ fall_4       <int> 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ price        <int> 0, 1, 0, 1, 0, 0, 0, 1, 1, 1, 0, 1, 1, NA, 1, 0, 1~
## $ body_part_1  <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1,~
## $ body_part_2  <int> 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 0, 0,~
## $ body_part_3  <int> 1, 0, 1, 1, 1, 0, 1, 1, 0, 0, 1, 1, 0, 0, 0, 0, 0,~
## $ body_part_4  <int> 1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ body_part_5  <int> 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ body_part_6  <int> 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0,~
## $ body_part_7  <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ body_part_8  <int> 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ body_part_9  <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ body_part_10 <int> 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0,~
## $ body_part_11 <int> 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0,~
## $ body_part_12 <int> 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0,~
## $ body_part_13 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0,~
## $ incentive    <int> 3, 1, 3, 3, 4, 3, 2, 3, 1, 1, 3, 2, 1, NA, 2, 2, 1~
## $ reasons_against_1 <int> 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0,~
## $ reasons_against_2 <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0,~
```

```
## $ reasons_against_3 <int> 0, 1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, ~
## $ reasons_against_4 <int> 1, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1, ~
## $ reasons_against_5 <int> 1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ reasons_against_6 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ diversity <int> 2, 2, 2, 2, 3, 2, 2, 2, 2, 1, 2, 2, 2, NA, 2, 2, 2, ~
## $ winter_apply_freq <int> 1, 3, 2, 4, 1, 1, 3, 3, 1, 3, 1, 3, 3, NA, 1, 3, 2, ~
## $ spring_apply_freq <int> 1, 2, 1, 4, 1, 1, 2, 2, 1, 2, 1, 3, 2, NA, 1, 2, 2, ~
## $ summer_apply_freq <int> 1, 2, 1, 4, 1, 1, 2, 2, 1, 2, 1, 4, 1, NA, 1, 1, 1, ~
## $ fall_apply_freq <int> 1, 3, 1, 4, 1, 1, 4, 3, 1, 3, 1, 3, 1, NA, 1, 2, 2, ~
```

```
recode_season_apply_freq <- function(x){
  dplyr::recode(x,
    `1` = 4L, # Yes = 1
    `2` = 3L,
    `3` = 1L,
    `4` = 2L,
    .default = NA_integer_)
}
```

```
data_with_seasons_recoded <- data_with_seasons %>%
  mutate(
    # recode
    winter_apply_freq = recode_season_apply_freq(winter_apply_freq),
    spring_apply_freq = recode_season_apply_freq(spring_apply_freq),
    summer_apply_freq = recode_season_apply_freq(summer_apply_freq),
    fall_apply_freq = recode_season_apply_freq(fall_apply_freq)
  )
```

```
glimpse(data_with_seasons_recoded)
```

```
## Rows: 1,008
## Columns: 58
## $ school <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, NA, 3, 1, 3~
## $ gender <int> 2, 2, 2, 2, 2, 2, 2, 2, 1, 2, 2, 2, 1, NA, 2, 2, N~
## $ seen_free_dispenser <int> 1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 0, 0, 1, NA, 0, 1, 0~
## $ would_you_use <int> 2, 2, 3, 2, 1, 1, 3, 3, 3, 3, 3, 1, 3, NA, 3, 3, 3~
## $ aid_in_prevention <int> 3, 3, 3, 2, 2, 3, 2, 3, 3, 3, 3, 3, 3, NA, 3, 3, 3~
## $ where_1 <int> 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 1, 0, ~
## $ where_2 <int> 1, 1, 1, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 0, ~
## $ where_3 <int> 0, 1, 0, 0, 1, 0, 1, 1, 0, 1, 1, 1, 0, 0, 0, 0, 0, ~
## $ where_4 <int> 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, ~
## $ where_5 <int> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 0, 0, 0, ~
## $ where_6 <int> 1, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, ~
## $ where_7 <int> 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 1, ~
## $ where_8 <int> 0, 1, 1, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0, 0, 0, 1, 0, ~
## $ where_9 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ skin_type <int> 3, 2, 3, 3, 4, 1, 4, 4, 3, 2, 1, 2, 2, NA, 3, 5, 1, ~
## $ apply_freq <int> 2, 2, 2, 4, 2, 2, 4, 3, 2, 2, 2, 1, 1, NA, 2, 2, N~
## $ winter_1 <int> 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 0, ~
## $ winter_2 <int> 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, ~
## $ winter_3 <int> 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, ~
## $ winter_4 <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ spring_1 <int> 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 0, ~
```

```
## $ spring_2      <int> 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 1, 0, 0, 1, 1, ~
## $ spring_3      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, ~
## $ spring_4      <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ summer_1      <int> 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0, 1, 0, 1, 1, 1, ~
## $ summer_2      <int> 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, ~
## $ summer_3      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ summer_4      <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, ~
## $ fall_1        <int> 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0, 1, 0, 1, 0, 0, ~
## $ fall_2        <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, ~
## $ fall_3        <int> 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, ~
## $ fall_4        <int> 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ price         <int> 0, 1, 0, 1, 0, 0, 0, 1, 1, 1, 0, 1, 1, NA, 1, 0, 1, ~
## $ body_part_1   <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, ~
## $ body_part_2   <int> 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 0, 0, ~
## $ body_part_3   <int> 1, 0, 1, 1, 1, 0, 1, 1, 0, 0, 1, 1, 0, 0, 0, 0, 0, ~
## $ body_part_4   <int> 1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ body_part_5   <int> 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, ~
## $ body_part_6   <int> 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, ~
## $ body_part_7   <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, ~
## $ body_part_8   <int> 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, ~
## $ body_part_9   <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, ~
## $ body_part_10  <int> 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, ~
## $ body_part_11  <int> 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, ~
## $ body_part_12  <int> 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, ~
## $ body_part_13  <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, ~
## $ incentive     <int> 3, 1, 3, 3, 4, 3, 2, 3, 1, 1, 3, 2, 1, NA, 2, 2, 1, ~
## $ reasons_against_1 <int> 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, ~
## $ reasons_against_2 <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, ~
## $ reasons_against_3 <int> 0, 1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, ~
## $ reasons_against_4 <int> 1, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1, ~
## $ reasons_against_5 <int> 1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ reasons_against_6 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ diversity     <int> 2, 2, 2, 2, 3, 2, 2, 2, 2, 1, 2, 2, 2, NA, 2, 2, 2, ~
## $ winter_apply_freq <int> 4, 1, 3, 2, 4, 4, 1, 1, 4, 1, 4, 1, 1, NA, 4, 1, 3, ~
## $ spring_apply_freq <int> 4, 3, 4, 2, 4, 4, 3, 3, 4, 3, 4, 1, 3, NA, 4, 3, 3, ~
## $ summer_apply_freq <int> 4, 3, 4, 2, 4, 4, 3, 3, 4, 3, 4, 2, 4, NA, 4, 4, 4, ~
## $ fall_apply_freq  <int> 4, 1, 4, 2, 4, 4, 2, 1, 4, 1, 4, 1, 4, NA, 4, 3, 3, ~
```

Remove the binary season application columns

```
season_onehots <- c(
  paste0("winter_", 1:4),
  paste0("spring_", 1:4),
  paste0("summer_", 1:4),
  paste0("fall_", 1:4)
)

data_final <- data_with_seasons_recoded %>%
  dplyr::select(-all_of(season_onehots))

# check
glimpse(data_final)
```

```
## Rows: 1,008
```

```
## Columns: 42
## $ school      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, NA, 3, 1, 3~
## $ gender      <int> 2, 2, 2, 2, 2, 2, 2, 2, 1, 2, 2, 2, 1, NA, 2, 2, N~
## $ seen_free_dispenser <int> 1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 0, 0, 1, NA, 0, 1, 0~
## $ would_you_use <int> 2, 2, 3, 2, 1, 1, 3, 3, 3, 3, 3, 1, 3, NA, 3, 3, 3~
## $ aid_in_prevention <int> 3, 3, 3, 2, 2, 3, 2, 3, 3, 3, 3, 3, 3, NA, 3, 3, 3~
## $ where_1      <int> 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 1, 0,~
## $ where_2      <int> 1, 1, 1, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 0,~
## $ where_3      <int> 0, 1, 0, 0, 1, 0, 1, 1, 0, 1, 1, 1, 0, 0, 0, 0, 0,~
## $ where_4      <int> 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ where_5      <int> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0,~
## $ where_6      <int> 1, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0,~
## $ where_7      <int> 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 1,~
## $ where_8      <int> 0, 1, 1, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0, 0, 0, 1, 0,~
## $ where_9      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ skin_type    <int> 3, 2, 3, 3, 4, 1, 4, 4, 3, 2, 1, 2, 2, NA, 3, 5, 1~
## $ apply_freq   <int> 2, 2, 2, 4, 2, 2, 4, 3, 2, 2, 2, 1, 1, NA, 2, 2, N~
## $ price        <int> 0, 1, 0, 1, 0, 0, 0, 1, 1, 1, 0, 1, 1, NA, 1, 0, 1~
## $ body_part_1  <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1,~
## $ body_part_2  <int> 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 0, 0,~
## $ body_part_3  <int> 1, 0, 1, 1, 1, 0, 1, 1, 0, 0, 1, 1, 0, 0, 0, 0, 0,~
## $ body_part_4  <int> 1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ body_part_5  <int> 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ body_part_6  <int> 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0,~
## $ body_part_7  <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ body_part_8  <int> 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ body_part_9  <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ body_part_10 <int> 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0,~
## $ body_part_11 <int> 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0,~
## $ body_part_12 <int> 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0,~
## $ body_part_13 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0,~
## $ incentive    <int> 3, 1, 3, 3, 4, 3, 2, 3, 1, 1, 3, 2, 1, NA, 2, 2, 1~
## $ reasons_against_1 <int> 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0,~
## $ reasons_against_2 <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0,~
## $ reasons_against_3 <int> 0, 1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0,~
## $ reasons_against_4 <int> 1, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1,~
## $ reasons_against_5 <int> 1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ reasons_against_6 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ diversity    <int> 2, 2, 2, 2, 3, 2, 2, 2, 2, 1, 2, 2, 2, NA, 2, 2, 2~
## $ winter_apply_freq <int> 4, 1, 3, 2, 4, 4, 1, 1, 4, 1, 4, 1, 1, NA, 4, 1, 3~
## $ spring_apply_freq <int> 4, 3, 4, 2, 4, 4, 3, 3, 4, 3, 4, 1, 3, NA, 4, 3, 3~
## $ summer_apply_freq <int> 4, 3, 4, 2, 4, 4, 3, 3, 4, 3, 4, 2, 4, NA, 4, 4, 4~
## $ fall_apply_freq <int> 4, 1, 4, 2, 4, 4, 2, 1, 4, 1, 4, 1, 4, NA, 4, 3, 3~
```

Create functions to match numeric row values with their survey labeled values

```
# Create Labels
# 1. School at Stony Brook University
label_school <- function(x){
  factor(x, levels = 1:3,
    labels = c("Undergraduate student",
               "Graduate student",
               "Other"))
}
```

```

# 2. Gender
label_gender <- function(x){
  factor(x, levels = 1:6,
    labels = c("Male", "Female", "Gender expansive",
      "Nonbinary", "Other", "Prefer not to say"))
}

# 3. Seen Free Sunscreen Dispenser (binary)
label_seen_free_dispenser <- function(x){
  factor(x, levels = 0:1,
    labels = c("No", "Yes"))
}

# 4. Would You Use Free Sunscreen Dispenser (Yes/No/Maybe)
label_would_you_use <- function(x){
  factor(x, levels = 1:3, labels = c("No", "Maybe", "Yes"), ordered = TRUE)
}

# 5. Aid in Prevention & Detection (Yes/No/Maybe)
label_aid_in_prevention <- function(x){
  factor(x, levels = 1:3, labels = c("No", "Maybe", "Yes"), ordered = TRUE)
}

# 6. Where Would You Use a Free Dispenser (multi-binary: 1=selected, 0=not)
label_where_dispenser <- c(
  "Exits/Entrances",
  "Bathrooms",
  "Bus stops",
  "Parking lots",
  "Classrooms",
  "Student health",
  "Student lounge",
  "Athletic fields",
  "Other Location"
)

# 7. Skin Type
label_skin <- function(x){
  factor(x, levels = 1:6,
    labels = c("I: Pale white",
      "II: Fair skin",
      "III: Darker white skin",
      "IV: Light brown skin",
      "V: Brown skin",
      "VI: Dark brown/black skin"),
    ordered = TRUE)
}

# 8. Overall Sunscreen Application Frequency
label_apply_freq <- function(x){
  factor(x, levels = 1:6, ordered = TRUE,
    labels = c("More than 1 time/day",
      "1 time/day",

```

```

        "1-3 times/week",
        "1-3 times/month",
        "1-3 times/year",
        "Never"))
}

# 9. Seasonal Sunscreen Application Frequency (collapsed ordinal)
label_season_freq <- function(x){
  factor(x, levels = 1:4, ordered = TRUE,
        labels = c("Never/Almost never",
                    "Specific outdoor occasions",
                    "Frequently/Occasionally",
                    "Always/Almost always"))
}

# 10. Price Barrier
label_price <- function(x){
  factor(x, levels = 0:1,
        labels = c("No", "Yes"))
}

# 11. Body Parts (multi-binary: 1=selected, 0=not)
label_body_parts <- c(
  "Face", "Head", "Neck", "Lips", "Ears", "Chest", "Abdomen",
  "Upper back", "Lower back", "Arms", "Hands", "Legs", "Feet"
)

# 12. Incentive for Sunscreen Use
label_incentive <- function(x){
  factor(x, levels = 1:4,
        labels = c("Prevent sunburn",
                    "Prevent skin cancer",
                    "Prevent wrinkles",
                    "Other Incentive"))
}

# 13. Reasons Against Applying / Reapplying Sunscreen (multi-binary)
label_reasons_against <- c(
  "Feels unpleasant",
  "Worried about skin reaction",
  "Cost",
  "Forget",
  "Feels unnecessary",
  "Other Reason"
)

# 14. Knowledge Question
label_diversity <- function(x){
  factor(x, levels = 1:3,
        labels = c("Correct", "Incorrect", "Unsure"))
}

```

Create labeled columns

```

# Create labeled dataset based on existing data_clean
data_labeled <- data_final %>%
  mutate(
    # Apply labeling functions to create "_lab" columns
    school_lab      = label_school(school),
    gender_lab      = label_gender(gender),
    seen_free_dispenser_lab = label_seen_free_dispenser(seen_free_dispenser),
    would_you_use_lab = label_would_you_use(would_you_use),
    aid_in_prevention_lab = label_aid_in_prevention(aid_in_prevention),
    skin_type_lab   = label_skin(skin_type),
    apply_freq_lab  = label_apply_freq(apply_freq),
    price_lab       = label_price(price),
    incentive_lab   = label_incentive(incentive),
    diversity_lab   = label_diversity(diversity),

    # Seasonal sunscreen frequency collapsed columns
    winter_apply_freq_lab = label_season_freq(winter_apply_freq),
    spring_apply_freq_lab = label_season_freq(spring_apply_freq),
    summer_apply_freq_lab = label_season_freq(summer_apply_freq),
    fall_apply_freq_lab   = label_season_freq(fall_apply_freq)
  )

glimpse(data_labeled)

```

```

## Rows: 1,008
## Columns: 56
## $ school      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, NA, 3, ~
## $ gender      <int> 2, 2, 2, 2, 2, 2, 2, 2, 1, 2, 2, 2, 1, NA, 2, ~
## $ seen_free_dispenser <int> 1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 0, 0, 1, NA, 0, ~
## $ would_you_use <int> 2, 2, 3, 2, 1, 1, 3, 3, 3, 3, 3, 1, 3, NA, 3, ~
## $ aid_in_prevention <int> 3, 3, 3, 2, 2, 3, 2, 3, 3, 3, 3, 3, 3, NA, 3, ~
## $ where_1      <int> 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 1~
## $ where_2      <int> 1, 1, 1, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1~
## $ where_3      <int> 0, 1, 0, 0, 1, 0, 1, 1, 0, 1, 1, 1, 0, 0, 0, 0~
## $ where_4      <int> 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0~
## $ where_5      <int> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0~
## $ where_6      <int> 1, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1~
## $ where_7      <int> 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0~
## $ where_8      <int> 0, 1, 1, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0, 0, 0, 1~
## $ where_9      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ skin_type    <int> 3, 2, 3, 3, 4, 1, 4, 4, 3, 2, 1, 2, 2, NA, 3, ~
## $ apply_freq   <int> 2, 2, 2, 4, 2, 2, 4, 3, 2, 2, 2, 1, 1, NA, 2, ~
## $ price        <int> 0, 1, 0, 1, 0, 0, 0, 1, 1, 1, 0, 1, 1, NA, 1, ~
## $ body_part_1  <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1~
## $ body_part_2  <int> 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 0~
## $ body_part_3  <int> 1, 0, 1, 1, 1, 0, 1, 1, 0, 0, 1, 1, 0, 0, 0, 0~
## $ body_part_4  <int> 1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0~
## $ body_part_5  <int> 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0~
## $ body_part_6  <int> 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0~
## $ body_part_7  <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0~
## $ body_part_8  <int> 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0~
## $ body_part_9  <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0~
## $ body_part_10 <int> 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 0, 0~

```

```
## $ body_part_11      <int> 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0~
## $ body_part_12      <int> 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 0, 0, 0~
## $ body_part_13      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0~
## $ incentive         <int> 3, 1, 3, 3, 4, 3, 2, 3, 1, 1, 3, 2, 1, NA, 2, ~
## $ reasons_against_1 <int> 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 0, 0~
## $ reasons_against_2 <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1~
## $ reasons_against_3 <int> 0, 1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0~
## $ reasons_against_4 <int> 1, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 1, 1~
## $ reasons_against_5 <int> 1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ reasons_against_6 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ diversity         <int> 2, 2, 2, 2, 3, 2, 2, 2, 2, 1, 2, 2, 2, NA, 2, ~
## $ winter_apply_freq <int> 4, 1, 3, 2, 4, 4, 1, 1, 4, 1, 4, 1, 1, NA, 4, ~
## $ spring_apply_freq <int> 4, 3, 4, 2, 4, 4, 3, 3, 4, 3, 4, 1, 3, NA, 4, ~
## $ summer_apply_freq <int> 4, 3, 4, 2, 4, 4, 3, 3, 4, 3, 4, 2, 4, NA, 4, ~
## $ fall_apply_freq   <int> 4, 1, 4, 2, 4, 4, 2, 1, 4, 1, 4, 1, 4, NA, 4, ~
## $ school_lab        <fct> Undergraduate student, Undergraduate student, ~
## $ gender_lab        <fct> Female, Female, Female, Female, Female, Female~
## $ seen_free_dispenser_lab <fct> Yes, No, No, No, Yes, No, No, Yes, Yes, No, No~
## $ would_you_use_lab  <ord> Maybe, Maybe, Yes, Maybe, No, No, Yes, Yes, Ye~
## $ aid_in_prevention_lab <ord> Yes, Yes, Yes, Maybe, Maybe, Yes, Maybe, Yes, ~
## $ skin_type_lab      <ord> III: Darker white skin, II: Fair skin, III: Da~
## $ apply_freq_lab     <ord> 1 time/day, 1 time/day, 1 time/day, 1-3 times/~
## $ price_lab          <fct> No, Yes, No, Yes, No, No, No, Yes, Yes, Yes, N~
## $ incentive_lab      <fct> Prevent wrinkles, Prevent sunburn, Prevent wri~
## $ diversity_lab      <fct> Incorrect, Incorrect, Incorrect, Incorrect, Un~
## $ winter_apply_freq_lab <ord> Always/Almost always, Never/Almost never, Freq~
## $ spring_apply_freq_lab <ord> Always/Almost always, Frequently/Occasionally,~
## $ summer_apply_freq_lab <ord> Always/Almost always, Frequently/Occasionally,~
## $ fall_apply_freq_lab <ord> Always/Almost always, Never/Almost never, Alwa~
```

Label the multinomial variable columns

```
# Rename WHERE columns
where_cols <- grep("^where_", colnames(data_labeled), value = TRUE)
colnames(data_labeled)[match(where_cols, colnames(data_labeled))] <- label_where_dispenser

# Rename BODY PART columns
body_cols <- grep("^body_part_", colnames(data_labeled), value = TRUE)
colnames(data_labeled)[match(body_cols, colnames(data_labeled))] <- label_body_parts

# Rename REASONS AGAINST columns
reason_cols <- grep("^reasons_against_", colnames(data_labeled), value = TRUE)
colnames(data_labeled)[match(reason_cols, colnames(data_labeled))] <- label_reasons_against

data_labeled <- data_labeled %>%
  dplyr::select(-matches("Other", ignore.case = TRUE))
```

```
glimpse(data_labeled)
```

```
## Rows: 1,008
## Columns: 54
## $ school      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, N~
## $ gender      <int> 2, 2, 2, 2, 2, 2, 2, 2, 1, 2, 2, 2, 1, N~
```

## \$ seen_free_dispenser	<int> 1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 0, 0, 1, N~
## \$ would_you_use	<int> 2, 2, 3, 2, 1, 1, 3, 3, 3, 3, 3, 1, 3, N~
## \$ aid_in_prevention	<int> 3, 3, 3, 2, 2, 3, 2, 3, 3, 3, 3, 3, 3, N~
## \$ 'Exits/Entrances'	<int> 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0~
## \$ Bathrooms	<int> 1, 1, 1, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1, 0~
## \$ 'Bus stops'	<int> 0, 1, 0, 0, 1, 0, 1, 1, 0, 1, 1, 1, 0, 0~
## \$ 'Parking lots'	<int> 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0~
## \$ Classrooms	<int> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0~
## \$ 'Student health'	<int> 1, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0~
## \$ 'Student lounge'	<int> 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0~
## \$ 'Athletic fields'	<int> 0, 1, 1, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0, 0~
## \$ skin_type	<int> 3, 2, 3, 3, 4, 1, 4, 4, 3, 2, 1, 2, 2, N~
## \$ apply_freq	<int> 2, 2, 2, 4, 2, 2, 4, 3, 2, 2, 2, 1, 1, N~
## \$ price	<int> 0, 1, 0, 1, 0, 0, 0, 1, 1, 1, 0, 1, 1, N~
## \$ Face	<int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0~
## \$ Head	<int> 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0~
## \$ Neck	<int> 1, 0, 1, 1, 1, 0, 1, 1, 0, 0, 1, 1, 0, 0~
## \$ Lips	<int> 1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0~
## \$ Ears	<int> 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0~
## \$ Chest	<int> 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0~
## \$ Abdomen	<int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0~
## \$ 'Upper back'	<int> 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0~
## \$ 'Lower back'	<int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0~
## \$ Arms	<int> 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0~
## \$ Hands	<int> 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0~
## \$ Legs	<int> 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 0~
## \$ Feet	<int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0~
## \$ incentive	<int> 3, 1, 3, 3, 4, 3, 2, 3, 1, 1, 3, 2, 1, N~
## \$ 'Feels unpleasant'	<int> 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0~
## \$ 'Worried about skin reaction'	<int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## \$ Cost	<int> 0, 1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0~
## \$ Forget	<int> 1, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0~
## \$ 'Feels unnecessary'	<int> 1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0~
## \$ diversity	<int> 2, 2, 2, 2, 3, 2, 2, 2, 2, 1, 2, 2, 2, N~
## \$ winter_apply_freq	<int> 4, 1, 3, 2, 4, 4, 1, 1, 4, 1, 4, 1, 1, N~
## \$ spring_apply_freq	<int> 4, 3, 4, 2, 4, 4, 3, 3, 4, 3, 4, 1, 3, N~
## \$ summer_apply_freq	<int> 4, 3, 4, 2, 4, 4, 3, 3, 4, 3, 4, 2, 4, N~
## \$ fall_apply_freq	<int> 4, 1, 4, 2, 4, 4, 2, 1, 4, 1, 4, 1, 4, N~
## \$ school_lab	<fct> Undergraduate student, Undergraduate stu~
## \$ gender_lab	<fct> Female, Female, Female, Female, Female, ~
## \$ seen_free_dispenser_lab	<fct> Yes, No, No, No, Yes, No, No, Yes, Yes, ~
## \$ would_you_use_lab	<ord> Maybe, Maybe, Yes, Maybe, No, No, Yes, Y~
## \$ aid_in_prevention_lab	<ord> Yes, Yes, Yes, Maybe, Maybe, Yes, Maybe,~
## \$ skin_type_lab	<ord> III: Darker white skin, II: Fair skin, I~
## \$ apply_freq_lab	<ord> 1 time/day, 1 time/day, 1 time/day, 1-3 ~
## \$ price_lab	<fct> No, Yes, No, Yes, No, No, No, Yes, Yes, ~
## \$ incentive_lab	<fct> Prevent wrinkles, Prevent sunburn, Preve~
## \$ diversity_lab	<fct> Incorrect, Incorrect, Incorrect, Incorre~
## \$ winter_apply_freq_lab	<ord> Always/Almost always, Never/Almost never~
## \$ spring_apply_freq_lab	<ord> Always/Almost always, Frequently/Occasio~
## \$ summer_apply_freq_lab	<ord> Always/Almost always, Frequently/Occasio~
## \$ fall_apply_freq_lab	<ord> Always/Almost always, Never/Almost never~

Collapse the body part variable categories into body regions

```
data_labeled <- data_labeled %>%
  mutate(
    body_headneck = if_any(c(Face, Head, Neck, Lips, Ears), ~ .x == 1),
    body_torso    = if_any(c(Chest, Abdomen, `Upper back`, `Lower back`), ~ .x == 1),
    body_limbs    = if_any(c(Arms, Hands, Legs, Feet), ~ .x == 1)
  ) %>%
  # convert TRUE/FALSE → 1/0
  mutate(across(starts_with("body_"), as.integer))
```

```
glimpse(data_labeled)
```

```
## Rows: 1,008
## Columns: 57
## $ school <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, N~
## $ gender <int> 2, 2, 2, 2, 2, 2, 2, 2, 1, 2, 2, 2, 1, N~
## $ seen_free_dispenser <int> 1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 0, 0, 1, N~
## $ would_you_use <int> 2, 2, 3, 2, 1, 1, 3, 3, 3, 3, 3, 1, 3, N~
## $ aid_in_prevention <int> 3, 3, 3, 2, 2, 3, 2, 3, 3, 3, 3, 3, 3, N~
## $ 'Exits/Entrances' <int> 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0~
## $ Bathrooms <int> 1, 1, 1, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1, 0~
## $ 'Bus stops' <int> 0, 1, 0, 0, 1, 0, 1, 1, 0, 1, 1, 1, 0, 0~
## $ 'Parking lots' <int> 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0~
## $ Classrooms <int> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0~
## $ 'Student health' <int> 1, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0~
## $ 'Student lounge' <int> 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0~
## $ 'Athletic fields' <int> 0, 1, 1, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0, 0~
## $ skin_type <int> 3, 2, 3, 3, 4, 1, 4, 4, 3, 2, 1, 2, 2, N~
## $ apply_freq <int> 2, 2, 2, 4, 2, 2, 4, 3, 2, 2, 2, 1, 1, N~
## $ price <int> 0, 1, 0, 1, 0, 0, 0, 1, 1, 1, 0, 1, 1, N~
## $ Face <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0~
## $ Head <int> 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0~
## $ Neck <int> 1, 0, 1, 1, 1, 0, 1, 1, 0, 0, 1, 1, 0, 0~
## $ Lips <int> 1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0~
## $ Ears <int> 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0~
## $ Chest <int> 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0~
## $ Abdomen <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0~
## $ 'Upper back' <int> 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0~
## $ 'Lower back' <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0~
## $ Arms <int> 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0~
## $ Hands <int> 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0~
## $ Legs <int> 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 0~
## $ Feet <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0~
## $ incentive <int> 3, 1, 3, 3, 4, 3, 2, 3, 1, 1, 3, 2, 1, N~
## $ 'Feels unpleasant' <int> 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0~
## $ 'Worried about skin reaction' <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ Cost <int> 0, 1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0~
## $ Forget <int> 1, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0~
## $ 'Feels unnecessary' <int> 1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0~
## $ diversity <int> 2, 2, 2, 2, 3, 2, 2, 2, 2, 1, 2, 2, 2, N~
## $ winter_apply_freq <int> 4, 1, 3, 2, 4, 4, 1, 1, 4, 1, 4, 1, 1, N~
## $ spring_apply_freq <int> 4, 3, 4, 2, 4, 4, 3, 3, 4, 3, 4, 1, 3, N~
## $ summer_apply_freq <int> 4, 3, 4, 2, 4, 4, 3, 3, 4, 3, 4, 2, 4, N~
## $ fall_apply_freq <int> 4, 1, 4, 2, 4, 4, 2, 1, 4, 1, 4, 1, 4, N~
```

```
## $ school_lab          <fct> Undergraduate student, Undergraduate stu-
## $ gender_lab          <fct> Female, Female, Female, Female, Female, ~
## $ seen_free_dispenser_lab <fct> Yes, No, No, No, Yes, No, No, Yes, Yes, ~
## $ would_you_use_lab    <ord> Maybe, Maybe, Yes, Maybe, No, No, Yes, Y-
## $ aid_in_prevention_lab <ord> Yes, Yes, Yes, Maybe, Maybe, Yes, Maybe,~
## $ skin_type_lab        <ord> III: Darker white skin, II: Fair skin, I-
## $ apply_freq_lab       <ord> 1 time/day, 1 time/day, 1 time/day, 1-3 ~
## $ price_lab           <fct> No, Yes, No, Yes, No, No, No, Yes, Yes, ~
## $ incentive_lab        <fct> Prevent wrinkles, Prevent sunburn, Preve-
## $ diversity_lab        <fct> Incorrect, Incorrect, Incorrect, Incorre-
## $ winter_apply_freq_lab <ord> Always/Almost always, Never/Almost never-
## $ spring_apply_freq_lab <ord> Always/Almost always, Frequently/Occasio-
## $ summer_apply_freq_lab <ord> Always/Almost always, Frequently/Occasio-
## $ fall_apply_freq_lab  <ord> Always/Almost always, Never/Almost never-
## $ body_headneck        <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0~
## $ body_torso           <int> 0, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 1, 1, 0~
## $ body_limbs           <int> 1, 0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0~
```

Check if labeled columns match the unlabeled ones

```
# confirm numeric label alignment
with(data_labeled, table(would_you_use, would_you_use_lab, useNA = "ifany"))
```

```
##           would_you_use_lab
## would_you_use  No Maybe Yes <NA>
##           1      91     0  0    0
##           2       0    199  0    0
##           3       0     0 446    0
##           <NA>    0     0  0  272
```

```
with(data_labeled, table(aid_in_prevention, aid_in_prevention_lab, useNA = "ifany"))
```

```
##           aid_in_prevention_lab
## aid_in_prevention  No Maybe Yes <NA>
##           1      38     0  0    0
##           2       0    84  0    0
##           3       0     0 615    0
##           <NA>    0     0  0  271
```

```
with(data_labeled, table(seen_free_dispenser, seen_free_dispenser_lab, useNA = "ifany"))
```

```
##           seen_free_dispenser_lab
## seen_free_dispenser  No Yes <NA>
##           0      475  0    0
##           1       0  260    0
##           <NA>    0   0  273
```

```
with(data_labeled, table(price, price_lab, useNA = "ifany"))
```

```
##           price_lab
## price  No Yes <NA>
```

```
##      0      325      0      0
##      1          0 411      0
##      <NA>      0      0 272
```

```
with(data_labeled, table(school, school_lab, useNA = "ifany"))
```

```
##      school_lab
## school Undergraduate student Graduate student Other <NA>
##      1              635              0      0      0
##      2              0              68      0      0
##      3              0              0      36      0
##      <NA>              0              0      0 269
```

```
with(data_labeled, table(gender, gender_lab, useNA = "ifany"))
```

```
##      gender_lab
## gender Male Female Gender expansive Nonbinary Other Prefer not to say <NA>
##      1      207      0              0      0      0              0      0
##      2          0 497              0      0      0              0      0
##      3          0      0              6      0      0              0      0
##      4          0      0              0      14      0              0      0
##      5          0      0              0      0      2              0      0
##      6          0      0              0      0      0              11      0
##      <NA>      0      0              0      0      0              0 271
```

```
with(data_labeled, table(skin_type, skin_type_lab, useNA = "ifany"))
```

```
##      skin_type_lab
## skin_type I: Pale white II: Fair skin III: Darker white skin
##      1              89              0              0
##      2              0              220              0
##      3              0              0              148
##      4              0              0              0
##      5              0              0              0
##      6              0              0              0
##      <NA>              0              0              0
##      skin_type_lab
## skin_type IV: Light brown skin V: Brown skin VI: Dark brown/black skin <NA>
##      1              0              0              0      0
##      2              0              0              0      0
##      3              0              0              0      0
##      4              166              0              0      0
##      5              0              92              0      0
##      6              0              0              24      0
##      <NA>              0              0              0 269
```

```
with(data_labeled, table(apply_freq, apply_freq_lab, useNA = "ifany"))
```

```
##      apply_freq_lab
## apply_freq More than 1 time/day 1 time/day 1-3 times/week 1-3 times/month
##      1              132              0              0              0
```

```
##           2           0           310           0           0
##           3           0           0           114          0
##           4           0           0           0           64
##           5           0           0           0           0
##           6           0           0           0           0
##          <NA>          0           0           0           0
##           apply_freq_lab
## apply_freq 1-3 times/year Never <NA>
##           1           0           0           0
##           2           0           0           0
##           3           0           0           0
##           4           0           0           0
##           5           61          0           0
##           6           0          53          0
##          <NA>          0           0          274
```

```
with(data_labeled, table(incentive, incentive_lab, useNA = "ifany"))
```

```
##           incentive_lab
## incentive Prevent sunburn Prevent skin cancer Prevent wrinkles Other Incentive
##           1           324           0           0           0
##           2           0           296           0           0
##           3           0           0           106          0
##           4           0           0           0           9
##          <NA>          0           0           0           0
##           incentive_lab
## incentive <NA>
##           1           0
##           2           0
##           3           0
##           4           0
##          <NA>          273
```

```
with(data_labeled, table(diversity, diversity_lab, useNA = "ifany"))
```

```
##           diversity_lab
## diversity Correct Incorrect Unsure <NA>
##           1           139           0           0           0
##           2           0           550           0           0
##           3           0           0           48          0
##          <NA>          0           0           0          271
```

```
with(data_labeled, table(winter_apply_freq, winter_apply_freq_lab, useNA = "ifany"))
```

```
##           winter_apply_freq_lab
## winter_apply_freq Never/Almost never Specific outdoor occasions
##           1           340           0
##           2           0           64
##           3           0           0
##           4           0           0
##          <NA>          0           0
##           winter_apply_freq_lab
```

```
## winter_apply_freq Frequently/Occasionally Always/Almost always <NA>
##           1                0                0    0
##           2                0                0    0
##           3               109                0    0
##           4                0               205    0
##           <NA>                0                0  290
```

```
with(data_labeled, table(spring_apply_freq, spring_apply_freq_lab, useNA = "ifany"))
```

```
##           spring_apply_freq_lab
## spring_apply_freq Never/Almost never Specific outdoor occasions
##           1               138                0
##           2                0               87
##           3                0                0
##           4                0                0
##           <NA>                0                0
##           spring_apply_freq_lab
## spring_apply_freq Frequently/Occasionally Always/Almost always <NA>
##           1                0                0    0
##           2                0                0    0
##           3               230                0    0
##           4                0               251    0
##           <NA>                0                0  302
```

```
with(data_labeled, table(summer_apply_freq, summer_apply_freq_lab, useNA = "ifany"))
```

```
##           summer_apply_freq_lab
## summer_apply_freq Never/Almost never Specific outdoor occasions
##           1                51                0
##           2                0               71
##           3                0                0
##           4                0                0
##           <NA>                0                0
##           summer_apply_freq_lab
## summer_apply_freq Frequently/Occasionally Always/Almost always <NA>
##           1                0                0    0
##           2                0                0    0
##           3               191                0    0
##           4                0               393    0
##           <NA>                0                0  302
```

```
with(data_labeled, table(fall_apply_freq, fall_apply_freq_lab, useNA = "ifany"))
```

```
##           fall_apply_freq_lab
## fall_apply_freq Never/Almost never Specific outdoor occasions
##           1               257                0
##           2                0               68
##           3                0                0
##           4                0                0
##           <NA>                0                0
##           fall_apply_freq_lab
## fall_apply_freq Frequently/Occasionally Always/Almost always <NA>
```

```
##           1           0           0   0
##           2           0           0   0
##           3          163           0   0
##           4           0          220   0
##          <NA>           0           0  300
```

We now have a cleaned data frame with both numeric and categorical columns that match.

2. Exploratory Data Analysis

Check collinearity

```
# Build the full "max model" including all major predictors except location columns
max_model <- glm(
  price_lab ~ school_lab + gender_lab + seen_free_dispenser_lab +
    would_you_use_lab + aid_in_prevention_lab + skin_type_lab +
    apply_freq_lab + winter_apply_freq_lab + spring_apply_freq_lab +
    summer_apply_freq_lab + fall_apply_freq_lab +
    body_headneck + body_torso + body_limbs +
    incentive_lab + diversity_lab +
    `Feels unpleasant` + `Worried about skin reaction` +
    Cost + Forget + `Feels unnecessary`,
  data = data_labeled,
  family = binomial
)

# Check VIFs
vif(max_model)
```

```
##           GVIF Df GVIF^(1/(2*Df))
## school_lab      1.201479  2      1.046957
## gender_lab      1.718523  5      1.055639
## seen_free_dispenser_lab 1.158282  1      1.076235
## would_you_use_lab 1.426766  2      1.092920
## aid_in_prevention_lab 1.448039  2      1.096971
## skin_type_lab    1.597343  5      1.047948
## apply_freq_lab   3.378568  5      1.129466
## winter_apply_freq_lab 14.006959  3      1.552592
## spring_apply_freq_lab 17.152476  3      1.605910
## summer_apply_freq_lab  6.790846  3      1.376113
## fall_apply_freq_lab 20.886864  3      1.659506
## body_headneck    1.080535  1      1.039488
## body_torso       1.376454  1      1.173224
## body_limbs       1.519254  1      1.232580
## incentive_lab    1.329699  3      1.048638
## diversity_lab    1.389524  2      1.085717
## `Feels unpleasant` 1.316981  1      1.147598
## `Worried about skin reaction` 1.251715  1      1.118801
## Cost            1.102272  1      1.049892
## Forget          1.613586  1      1.270270
## `Feels unnecessary` 1.238916  1      1.113066
```

3. Modeling

```
model_applyfreq <- polr(apply_freq_lab ~ gender_lab + skin_type_lab + incentive_lab +
  price_lab + `Feels unpleasant` + `Worried about skin reaction` + Cost + Forget +
  `Feels unnecessary`, data = data_labeled, Hess = TRUE)
```

```
# Extract coefficients
(ctable <- coef(summary(model_applyfreq)))
```

##	Value	Std. Error	t value
## gender_labFemale	-0.66432006	0.1639117	-4.0529144
## gender_labGender expansive	-0.90484018	0.9011840	-1.0040571
## gender_labNonbinary	1.56605616	0.4865943	3.2184020
## gender_labOther	-2.61867124	1.3770624	-1.9016359
## gender_labPrefer not to say	-0.07661714	0.5714858	-0.1340666
## skin_type_lab.L	0.39289202	0.2748630	1.4294106
## skin_type_lab.Q	-0.16111567	0.2561867	-0.6288993
## skin_type_lab.C	-0.25902141	0.2138160	-1.2114221
## skin_type_lab^4	-0.21960983	0.1754744	-1.2515206
## skin_type_lab^5	0.13326953	0.1563516	0.8523705
## incentive_labPrevent skin cancer	-0.26459565	0.1535512	-1.7231758
## incentive_labPrevent wrinkles	-0.19320364	0.2042326	-0.9459979
## incentive_labOther Incentive	-0.41344690	0.6086404	-0.6792959
## price_labYes	-0.28173032	0.1510053	-1.8656986
## `Feels unpleasant`	0.52506131	0.1613623	3.2539279
## `Worried about skin reaction`	0.25241994	0.1930051	1.3078407
## Cost	0.12087792	0.1719555	0.7029603
## Forget	0.15515982	0.1547294	1.0027818
## `Feels unnecessary`	0.96414608	0.2184791	4.4129891
## More than 1 time/day 1 time/day	-1.99855768	0.2362209	-8.4605442
## 1 time/day 1-3 times/week	0.07164775	0.2242981	0.3194309
## 1-3 times/week 1-3 times/month	0.88242201	0.2262332	3.9004979
## 1-3 times/month 1-3 times/year	1.52863394	0.2327654	6.5672742
## 1-3 times/year Never	2.46480543	0.2545805	9.6818324

```
# Raw 2-sided p-values
p_values <- 2 * pnorm(abs(ctable[, "t value"]), lower.tail = FALSE)

# Adjust them (choose method: "BH" = FDR, "holm" = step-down, "bonferroni" = strictest)
p_values_adj <- p.adjust(p_values, method = "BH")

# Bind both to the coefficient table
ctable <- cbind(ctable,
  "p value" = formatC(p_values, format = "f", digits = 4),
  "adj p (BH)" = formatC(p_values_adj, format = "f", digits = 4))

ctable
```

##	Value	Std. Error
## gender_labFemale	"-0.664320060421677"	"0.163911692894065"
## gender_labGender expansive	"-0.904840175167459"	"0.901183993439321"
## gender_labNonbinary	"1.56605616216173"	"0.486594332443145"

## gender_labOther	"-2.61867123776808"	"1.37706237201972"
## gender_labPrefer not to say	"-0.076617142635305"	"0.571485845511004"
## skin_type_lab.L	"0.392892024393372"	"0.274862961146285"
## skin_type_lab.Q	"-0.161115666075951"	"0.2561867407681"
## skin_type_lab.C	"-0.259021409647109"	"0.21381598342075"
## skin_type_lab^4	"-0.219609825892265"	"0.175474395527385"
## skin_type_lab^5	"0.133269530085077"	"0.156351636580948"
## incentive_labPrevent skin cancer	"-0.264595654467567"	"0.153551169010648"
## incentive_labPrevent wrinkles	"-0.193203641269194"	"0.204232642832706"
## incentive_labOther Incentive	"-0.413446895141255"	"0.608640368634836"
## price_labYes	"-0.28173031967992"	"0.151005267289138"
## 'Feels unpleasant'	"0.525061305342577"	"0.161362304423445"
## 'Worried about skin reaction'	"0.252419939180226"	"0.193005111121508"
## Cost	"0.12087792313055"	"0.171955546855244"
## Forget	"0.155159822749931"	"0.154729401276553"
## 'Feels unnecessary'	"0.96414608484333"	"0.218479145305331"
## More than 1 time/day 1 time/day	"-1.99855768445813"	"0.236220937112766"
## 1 time/day 1-3 times/week	"0.0716477490171525"	"0.224298101580176"
## 1-3 times/week 1-3 times/month	"0.88242200687374"	"0.226233173476167"
## 1-3 times/month 1-3 times/year	"1.52863394325014"	"0.232765361745402"
## 1-3 times/year Never	"2.4648054331737"	"0.254580468096852"
##	t value	p value adj p (BH)
## gender_labFemale	"-4.05291439977391"	"0.0001" "0.0002"
## gender_labGender expansive	"-1.00405708684881"	"0.3154" "0.4461"
## gender_labNonbinary	"3.21840197829413"	"0.0013" "0.0039"
## gender_labOther	"-1.90163589607586"	"0.0572" "0.1490"
## gender_labPrefer not to say	"-0.134066562167951"	"0.8933" "0.8933"
## skin_type_lab.L	"1.42941057883849"	"0.1529" "0.3058"
## skin_type_lab.Q	"-0.628899316150764"	"0.5294" "0.5775"
## skin_type_lab.C	"-1.21142210934439"	"0.2257" "0.3612"
## skin_type_lab^4	"-1.25152062916206"	"0.2107" "0.3612"
## skin_type_lab^5	"0.852370547564302"	"0.3940" "0.4977"
## incentive_labPrevent skin cancer	"-1.72317577373324"	"0.0849" "0.1851"
## incentive_labPrevent wrinkles	"-0.945997851222315"	"0.3441" "0.4589"
## incentive_labOther Incentive	"-0.679295880535504"	"0.4970" "0.5679"
## price_labYes	"-1.86569862586632"	"0.0621" "0.1490"
## 'Feels unpleasant'	"3.2539279060165"	"0.0011" "0.0039"
## 'Worried about skin reaction'	"1.30784069765548"	"0.1909" "0.3525"
## Cost	"0.702960301898882"	"0.4821" "0.5679"
## Forget	"1.00278176913907"	"0.3160" "0.4461"
## 'Feels unnecessary'	"4.41298909099956"	"0.0000" "0.0001"
## More than 1 time/day 1 time/day	"-8.46054422137892"	"0.0000" "0.0000"
## 1 time/day 1-3 times/week	"0.319430920335016"	"0.7494" "0.7820"
## 1-3 times/week 1-3 times/month	"3.90049785058025"	"0.0001" "0.0004"
## 1-3 times/month 1-3 times/year	"6.5672741501897"	"0.0000" "0.0000"
## 1-3 times/year Never	"9.68183243435626"	"0.0000" "0.0000"

```
# Load library
```

```
library(nnet)
```

```
# Fit multinomial logistic regression
```

```
model_incentive <- multinom(
```

```
  incentive_lab ~ winter_apply_freq_lab + spring_apply_freq_lab +  
    summer_apply_freq_lab + fall_apply_freq_lab +
```

```

    gender_lab + skin_type_lab + price_lab +
    body_headneck + body_torso + body_limbs,
    data = data_labeled
)

```

```

## # weights: 112 (81 variable)
## initial value 927.430928
## iter 10 value 711.367598
## iter 20 value 669.989640
## iter 30 value 661.224840
## iter 40 value 657.831037
## iter 50 value 656.952054
## iter 60 value 656.775182
## iter 70 value 656.677187
## iter 80 value 656.672713
## final value 656.672570
## converged

```

```

summary_mod <- summary(model_incentive)

```

```

# Extract coefficients & SEs
sm <- summary(model_incentive)
bet <- sm$coefficients           # matrix: rows = outcomes (vs baseline), cols = predictors
se <- sm$standard.errors        # same shape

```

```

# z, p (two-sided)
z <- bet / se
p <- 2 * (1 - pnorm(abs(z)))

```

```

# Build a long data frame similar to your polr ctable

```

```

ctable <- data.frame(
  Outcome = rep(rownames(bet), each = ncol(bet)),
  Predictor = rep(colnames(bet), times = nrow(bet)),
  Value = as.vector(t(bet)),
  `Std. Error` = as.vector(t(se)),
  `z value` = as.vector(t(z)),
  check.names = FALSE
)

```

```

# Add raw p and BH-adjusted p across all coefficients in the model

```

```

p_vec <- as.vector(t(p))
ctable$`p value` = formatC(p_vec, format = "f", digits = 4)
ctable$`adj p (BH)` = formatC(p.adjust(p_vec, method = "BH"), format = "f", digits = 4)

```

```

# (Optional) add Odds Ratios

```

```

ctable$`Odds Ratio` <- round(exp(ctable$Value), 3)

```

```

# Nice ordering

```

```

ctable <- ctable[, c("Outcome", "Predictor", "Value", "Std. Error", "z value", "Odds Ratio", "p value", "adj p
ctable

```

```

## Outcome Predictor Value Std. Error

```

## 1	Prevent skin cancer	(Intercept)	0.49390076	5.780290e-01
## 2	Prevent skin cancer	winter_apply_freq_lab.L	-0.18334402	3.618187e-01
## 3	Prevent skin cancer	winter_apply_freq_lab.Q	-0.45678760	2.879866e-01
## 4	Prevent skin cancer	winter_apply_freq_lab.C	-0.15838190	3.039263e-01
## 5	Prevent skin cancer	spring_apply_freq_lab.L	-0.39949230	3.691222e-01
## 6	Prevent skin cancer	spring_apply_freq_lab.Q	-0.40617266	2.766383e-01
## 7	Prevent skin cancer	spring_apply_freq_lab.C	0.44243990	2.642702e-01
## 8	Prevent skin cancer	summer_apply_freq_lab.L	0.29783266	3.222418e-01
## 9	Prevent skin cancer	summer_apply_freq_lab.Q	0.22696867	2.683795e-01
## 10	Prevent skin cancer	summer_apply_freq_lab.C	-0.25309769	2.613339e-01
## 11	Prevent skin cancer	fall_apply_freq_lab.L	0.75604205	3.817718e-01
## 12	Prevent skin cancer	fall_apply_freq_lab.Q	0.40840097	3.015676e-01
## 13	Prevent skin cancer	fall_apply_freq_lab.C	0.14259252	3.037895e-01
## 14	Prevent skin cancer	gender_labFemale	0.06792979	2.007534e-01
## 15	Prevent skin cancer	gender_labGender expansive	-1.38543224	1.157310e+00
## 16	Prevent skin cancer	gender_labNonbinary	-0.06269985	6.581220e-01
## 17	Prevent skin cancer	gender_labOther	-27.98871237	2.354209e-14
## 18	Prevent skin cancer	gender_labPrefer not to say	0.51786572	7.328516e-01
## 19	Prevent skin cancer	skin_type_lab.L	0.52723203	3.500733e-01
## 20	Prevent skin cancer	skin_type_lab.Q	-0.26877084	3.236295e-01
## 21	Prevent skin cancer	skin_type_lab.C	-0.43546556	2.719130e-01
## 22	Prevent skin cancer	skin_type_lab^4	0.07006390	2.303944e-01
## 23	Prevent skin cancer	skin_type_lab^5	-0.47304220	1.988745e-01
## 24	Prevent skin cancer	price_labYes	0.17353202	1.837782e-01
## 25	Prevent skin cancer	body_headneck	-0.71127657	5.293991e-01
## 26	Prevent skin cancer	body_torso	0.08008893	2.122010e-01
## 27	Prevent skin cancer	body_limbs	0.04388249	2.046897e-01
## 28	Prevent wrinkles	(Intercept)	-1.48964522	9.354681e-01
## 29	Prevent wrinkles	winter_apply_freq_lab.L	0.11855698	4.939848e-01
## 30	Prevent wrinkles	winter_apply_freq_lab.Q	-0.62436257	3.943529e-01
## 31	Prevent wrinkles	winter_apply_freq_lab.C	-0.35135309	4.144325e-01
## 32	Prevent wrinkles	spring_apply_freq_lab.L	0.03600977	5.383619e-01
## 33	Prevent wrinkles	spring_apply_freq_lab.Q	-0.43741525	4.210755e-01
## 34	Prevent wrinkles	spring_apply_freq_lab.C	0.07292340	4.352028e-01
## 35	Prevent wrinkles	summer_apply_freq_lab.L	0.41475929	5.476523e-01
## 36	Prevent wrinkles	summer_apply_freq_lab.Q	-0.05093824	4.744576e-01
## 37	Prevent wrinkles	summer_apply_freq_lab.C	0.04465981	4.604702e-01
## 38	Prevent wrinkles	fall_apply_freq_lab.L	0.85240051	5.241830e-01
## 39	Prevent wrinkles	fall_apply_freq_lab.Q	0.80332822	4.702528e-01
## 40	Prevent wrinkles	fall_apply_freq_lab.C	-0.45676824	5.244717e-01
## 41	Prevent wrinkles	gender_labFemale	0.54482985	3.129888e-01
## 42	Prevent wrinkles	gender_labGender expansive	-0.21430278	1.229172e+00
## 43	Prevent wrinkles	gender_labNonbinary	0.57478109	9.237663e-01
## 44	Prevent wrinkles	gender_labOther	-17.43836341	6.540412e-10
## 45	Prevent wrinkles	gender_labPrefer not to say	0.34177139	1.188878e+00
## 46	Prevent wrinkles	skin_type_lab.L	0.49280579	4.510140e-01
## 47	Prevent wrinkles	skin_type_lab.Q	0.15313056	4.190235e-01
## 48	Prevent wrinkles	skin_type_lab.C	-0.26365764	3.575959e-01
## 49	Prevent wrinkles	skin_type_lab^4	0.17225461	3.117156e-01
## 50	Prevent wrinkles	skin_type_lab^5	-0.64603303	2.818045e-01
## 51	Prevent wrinkles	price_labYes	-0.09271154	2.530389e-01
## 52	Prevent wrinkles	body_headneck	0.00586777	8.590458e-01
## 53	Prevent wrinkles	body_torso	0.04403468	3.074902e-01
## 54	Prevent wrinkles	body_limbs	-0.25886908	2.911061e-01

## 55	Other Incentive	(Intercept)	-53.35716756	4.917539e-01
## 56	Other Incentive	winter_apply_freq_lab.L	-5.35629983	2.012620e+00
## 57	Other Incentive	winter_apply_freq_lab.Q	13.86485051	1.281108e+00
## 58	Other Incentive	winter_apply_freq_lab.C	19.83808101	9.774581e-01
## 59	Other Incentive	spring_apply_freq_lab.L	0.53653700	2.073410e+00
## 60	Other Incentive	spring_apply_freq_lab.Q	0.68788870	1.760423e+00
## 61	Other Incentive	spring_apply_freq_lab.C	0.67405527	1.474005e+00
## 62	Other Incentive	summer_apply_freq_lab.L	4.41366449	2.062510e+00
## 63	Other Incentive	summer_apply_freq_lab.Q	15.68411832	1.051931e+00
## 64	Other Incentive	summer_apply_freq_lab.C	-23.10971035	1.193924e+00
## 65	Other Incentive	fall_apply_freq_lab.L	-4.95421803	2.006297e+00
## 66	Other Incentive	fall_apply_freq_lab.Q	13.20037872	1.163287e+00
## 67	Other Incentive	fall_apply_freq_lab.C	18.67274177	8.077160e-01
## 68	Other Incentive	gender_labFemale	0.24013920	9.810084e-01
## 69	Other Incentive	gender_labGender expansive	-4.10808927	1.146866e-15
## 70	Other Incentive	gender_labNonbinary	-17.29204300	1.864954e-08
## 71	Other Incentive	gender_labOther	-2.21153539	5.716557e-16
## 72	Other Incentive	gender_labPrefer not to say	-7.96370754	1.363697e-05
## 73	Other Incentive	skin_type_lab.L	8.07620836	4.531598e-01
## 74	Other Incentive	skin_type_lab.Q	-20.47429771	2.398037e-01
## 75	Other Incentive	skin_type_lab.C	-5.75577591	6.471515e-01
## 76	Other Incentive	skin_type_lab^4	-23.50658554	6.571170e-01
## 77	Other Incentive	skin_type_lab^5	19.63540274	7.035263e-01
## 78	Other Incentive	price_labYes	-1.15638174	9.113775e-01
## 79	Other Incentive	body_headneck	13.07631155	4.917539e-01
## 80	Other Incentive	body_torso	0.73231019	9.016920e-01
## 81	Other Incentive	body_limbs	1.28124229	9.752138e-01
##	z value	Odds Ratio	p value	adj p (BH)
## 1	8.544568e-01	1.639000e+00	0.3929	0.6317
## 2	-5.067290e-01	8.320000e-01	0.6123	0.8267
## 3	-1.586142e+00	6.330000e-01	0.1127	0.2783
## 4	-5.211194e-01	8.540000e-01	0.6023	0.8267
## 5	-1.082277e+00	6.710000e-01	0.2791	0.5383
## 6	-1.468244e+00	6.660000e-01	0.1420	0.3287
## 7	1.674195e+00	1.557000e+00	0.0941	0.2722
## 8	9.242520e-01	1.347000e+00	0.3554	0.6257
## 9	8.457004e-01	1.255000e+00	0.3977	0.6317
## 10	-9.684841e-01	7.760000e-01	0.3328	0.6127
## 11	1.980351e+00	2.130000e+00	0.0477	0.1544
## 12	1.354260e+00	1.504000e+00	0.1757	0.3921
## 13	4.693794e-01	1.153000e+00	0.6388	0.8459
## 14	3.383742e-01	1.070000e+00	0.7351	0.8887
## 15	-1.197114e+00	2.500000e-01	0.2313	0.4683
## 16	-9.527087e-02	9.390000e-01	0.9241	0.9475
## 17	-1.188880e+15	0.000000e+00	0.0000	0.0000
## 18	7.066447e-01	1.678000e+00	0.4798	0.6940
## 19	1.506062e+00	1.694000e+00	0.1321	0.3146
## 20	-8.304894e-01	7.640000e-01	0.4063	0.6328
## 21	-1.601489e+00	6.470000e-01	0.1093	0.2783
## 22	3.041042e-01	1.073000e+00	0.7610	0.9065
## 23	-2.378597e+00	6.230000e-01	0.0174	0.0640
## 24	9.442473e-01	1.189000e+00	0.3450	0.6211
## 25	-1.343555e+00	4.910000e-01	0.1791	0.3921
## 26	3.774201e-01	1.083000e+00	0.7059	0.8772

## 27	2.143854e-01	1.045000e+00	0.8302	0.9212
## 28	-1.592406e+00	2.250000e-01	0.1113	0.2783
## 29	2.400013e-01	1.126000e+00	0.8103	0.9116
## 30	-1.583259e+00	5.360000e-01	0.1134	0.2783
## 31	-8.477933e-01	7.040000e-01	0.3966	0.6317
## 32	6.688767e-02	1.037000e+00	0.9467	0.9585
## 33	-1.038805e+00	6.460000e-01	0.2989	0.5630
## 34	1.675619e-01	1.076000e+00	0.8669	0.9363
## 35	7.573406e-01	1.514000e+00	0.4488	0.6733
## 36	-1.073610e-01	9.500000e-01	0.9145	0.9475
## 37	9.698742e-02	1.046000e+00	0.9227	0.9475
## 38	1.626151e+00	2.345000e+00	0.1039	0.2783
## 39	1.708290e+00	2.233000e+00	0.0876	0.2627
## 40	-8.709111e-01	6.330000e-01	0.3838	0.6317
## 41	1.740733e+00	1.724000e+00	0.0817	0.2546
## 42	-1.743472e-01	8.070000e-01	0.8616	0.9363
## 43	6.222148e-01	1.777000e+00	0.5338	0.7586
## 44	-2.666248e+10	0.000000e+00	0.0000	0.0000
## 45	2.874738e-01	1.407000e+00	0.7737	0.9083
## 46	1.092662e+00	1.637000e+00	0.2745	0.5383
## 47	3.654462e-01	1.165000e+00	0.7148	0.8772
## 48	-7.373061e-01	7.680000e-01	0.4609	0.6788
## 49	5.526018e-01	1.188000e+00	0.5805	0.8107
## 50	-2.292486e+00	5.240000e-01	0.0219	0.0770
## 51	-3.663924e-01	9.110000e-01	0.7141	0.8772
## 52	6.830567e-03	1.006000e+00	0.9946	0.9946
## 53	1.432068e-01	1.045000e+00	0.8861	0.9444
## 54	-8.892603e-01	7.720000e-01	0.3739	0.6317
## 55	-1.085038e+02	0.000000e+00	0.0000	0.0000
## 56	-2.661356e+00	5.000000e-03	0.0078	0.0315
## 57	1.082254e+01	1.050577e+06	0.0000	0.0000
## 58	2.029558e+01	4.126379e+08	0.0000	0.0000
## 59	2.587703e-01	1.710000e+00	0.7958	0.9116
## 60	3.907520e-01	1.990000e+00	0.6960	0.8772
## 61	4.572950e-01	1.962000e+00	0.6475	0.8459
## 62	2.139948e+00	8.257100e+01	0.0324	0.1092
## 63	1.490984e+01	6.479269e+06	0.0000	0.0000
## 64	-1.935610e+01	0.000000e+00	0.0000	0.0000
## 65	-2.469334e+00	7.000000e-03	0.0135	0.0522
## 66	1.134748e+01	5.405696e+05	0.0000	0.0000
## 67	2.311796e+01	1.286675e+08	0.0000	0.0000
## 68	2.447881e-01	1.271000e+00	0.8066	0.9116
## 69	-3.582012e+15	1.600000e-02	0.0000	0.0000
## 70	-9.272100e+08	0.000000e+00	0.0000	0.0000
## 71	-3.868649e+15	1.100000e-01	0.0000	0.0000
## 72	-5.839791e+05	0.000000e+00	0.0000	0.0000
## 73	1.782199e+01	3.217012e+03	0.0000	0.0000
## 74	-8.537942e+01	0.000000e+00	0.0000	0.0000
## 75	-8.894016e+00	3.000000e-03	0.0000	0.0000
## 76	-3.577230e+01	0.000000e+00	0.0000	0.0000
## 77	2.790998e+01	3.369357e+08	0.0000	0.0000
## 78	-1.268828e+00	3.150000e-01	0.2045	0.4247
## 79	2.659117e+01	4.774962e+05	0.0000	0.0000
## 80	8.121511e-01	2.080000e+00	0.4167	0.6369

```
## 81 1.313807e+00 3.601000e+00 0.1889 0.4027
```

```
model_season <- polr(summer_apply_freq_lab ~ winter_apply_freq_lab +
                     spring_apply_freq_lab + fall_apply_freq_lab +
                     apply_freq_lab + gender_lab + skin_type_lab +
                     price_lab,
                     data = data_labeled, Hess = TRUE)
# Get coefficient summary
ctable <- coef(summary(model_season))

# Calculate raw two-sided p-values
p_values <- 2 * pnorm(abs(ctable[, "t value"]), lower.tail = FALSE)

# Benjamini-Hochberg correction for multiple testing
p_values_adj <- p.adjust(p_values, method = "BH")

# Bind raw + adjusted p-values into the table
ctable <- cbind(
  ctable,
  "p value" = formatC(p_values, format = "f", digits = 4),
  "adj p (BH)" = formatC(p_values_adj, format = "f", digits = 4)
)

# Final output table
ctable
```

##	Value
## winter_apply_freq_lab.L	"-3.36319288647434"
## winter_apply_freq_lab.Q	"-1.54395896372251"
## winter_apply_freq_lab.C	"0.11385499013849"
## spring_apply_freq_lab.L	"5.37821996558551"
## spring_apply_freq_lab.Q	"2.73798975326576"
## spring_apply_freq_lab.C	"0.318626750499599"
## fall_apply_freq_lab.L	"2.84648910083729"
## fall_apply_freq_lab.Q	"0.904021872269106"
## fall_apply_freq_lab.C	"0.245284007726813"
## apply_freq_lab.L	"-1.80528560773195"
## apply_freq_lab.Q	"-1.34818856019955"
## apply_freq_lab.C	"0.0767918474343979"
## apply_freq_lab^4	"-0.175450057154096"
## apply_freq_lab^5	"-0.468994121700642"
## gender_labFemale	"0.428399780822721"
## gender_labGender expansive	"0.238972833365402"
## gender_labNonbinary	"1.29079881689853"
## gender_labOther	"-0.792946188520121"
## gender_labPrefer not to say	"-0.669243289559342"
## skin_type_lab.L	"0.0877821474171521"
## skin_type_lab.Q	"-0.200059649754271"
## skin_type_lab.C	"0.464629492306047"
## skin_type_lab^4	"0.346333509795063"
## skin_type_lab^5	"-0.310065388335321"
## price_labYes	"-0.0339953963868826"
## Never/Almost never Specific outdoor occasions	"-3.75936647963523"
## Specific outdoor occasions Frequently/Occasionally	"-2.14993695186122"

## Frequently/Occasionally/Always/Almost always	"0.801732835356246"
##	Std. Error
## winter_apply_freq_lab.L	"0.515898990723119"
## winter_apply_freq_lab.Q	"0.400199632955468"
## winter_apply_freq_lab.C	"0.349487766918189"
## spring_apply_freq_lab.L	"0.7078328035149"
## spring_apply_freq_lab.Q	"0.528286544627734"
## spring_apply_freq_lab.C	"0.287430829453103"
## fall_apply_freq_lab.L	"0.575918013334318"
## fall_apply_freq_lab.Q	"0.429763275805661"
## fall_apply_freq_lab.C	"0.336631462211526"
## apply_freq_lab.L	"0.307380167024343"
## apply_freq_lab.Q	"0.280581043863648"
## apply_freq_lab.C	"0.252452725584879"
## apply_freq_lab^4	"0.237759684054376"
## apply_freq_lab^5	"0.24811292270694"
## gender_labFemale	"0.219890978743677"
## gender_labGender expansive	"0.791168351688215"
## gender_labNonbinary	"0.670881991888087"
## gender_labOther	"1.58936171623117"
## gender_labPrefer not to say	"0.715005725209552"
## skin_type_lab.L	"0.396078487161415"
## skin_type_lab.Q	"0.368541760490217"
## skin_type_lab.C	"0.303347363733701"
## skin_type_lab^4	"0.247620529203063"
## skin_type_lab^5	"0.223873810016523"
## price_labYes	"0.201842436994562"
## Never/Almost never/Specific outdoor occasions	"0.313840172629013"
## Specific outdoor occasions/Frequently/Occasionally	"0.284953906149034"
## Frequently/Occasionally/Always/Almost always	"0.28417424023797"
##	t value
## winter_apply_freq_lab.L	"-6.51909181245007"
## winter_apply_freq_lab.Q	"-3.85797196344333"
## winter_apply_freq_lab.C	"0.325776753625662"
## spring_apply_freq_lab.L	"7.5981502112911"
## spring_apply_freq_lab.Q	"5.18277397202144"
## spring_apply_freq_lab.C	"1.1085336639283"
## fall_apply_freq_lab.L	"4.94252486453295"
## fall_apply_freq_lab.Q	"2.10353448785118"
## fall_apply_freq_lab.C	"0.728642552052031"
## apply_freq_lab.L	"-5.87313627033385"
## apply_freq_lab.Q	"-4.80498804065582"
## apply_freq_lab.C	"0.3041830792537"
## apply_freq_lab^4	"-0.737930225016496"
## apply_freq_lab^5	"-1.8902446377394"
## gender_labFemale	"1.94823718221792"
## gender_labGender expansive	"0.302050546960677"
## gender_labNonbinary	"1.92403259068825"
## gender_labOther	"-0.49890857469527"
## gender_labPrefer not to say	"-0.935997105985692"
## skin_type_lab.L	"0.221628162756989"
## skin_type_lab.Q	"-0.542841195223468"
## skin_type_lab.C	"1.53167473284498"
## skin_type_lab^4	"1.39864619023995"

```
## skin_type_lab^5 "-1.38500072122075"
## price_labYes "-0.168425415849485"
## Never/Almost never|Specific outdoor occasions "-11.9786018728684"
## Specific outdoor occasions|Frequently/Occasionally "-7.54485867878149"
## Frequently/Occasionally|Always/Almost always "2.82127202903708"
## p value adj p (BH)
## winter_apply_freq_lab.L "0.0000" "0.0000"
## winter_apply_freq_lab.Q "0.0001" "0.0004"
## winter_apply_freq_lab.C "0.7446" "0.8213"
## spring_apply_freq_lab.L "0.0000" "0.0000"
## spring_apply_freq_lab.Q "0.0000" "0.0000"
## spring_apply_freq_lab.C "0.2676" "0.4163"
## fall_apply_freq_lab.L "0.0000" "0.0000"
## fall_apply_freq_lab.Q "0.0354" "0.0902"
## fall_apply_freq_lab.C "0.4662" "0.6216"
## apply_freq_lab.L "0.0000" "0.0000"
## apply_freq_lab.Q "0.0000" "0.0000"
## apply_freq_lab.C "0.7610" "0.8213"
## apply_freq_lab^4 "0.4606" "0.6216"
## apply_freq_lab^5 "0.0587" "0.1175"
## gender_labFemale "0.0514" "0.1171"
## gender_labGender expansive "0.7626" "0.8213"
## gender_labNonbinary "0.0544" "0.1171"
## gender_labOther "0.6178" "0.7522"
## gender_labPrefer not to say "0.3493" "0.5147"
## skin_type_lab.L "0.8246" "0.8551"
## skin_type_lab.Q "0.5872" "0.7474"
## skin_type_lab.C "0.1256" "0.2345"
## skin_type_lab^4 "0.1619" "0.2735"
## skin_type_lab^5 "0.1661" "0.2735"
## price_labYes "0.8662" "0.8662"
## Never/Almost never|Specific outdoor occasions "0.0000" "0.0000"
## Specific outdoor occasions|Frequently/Occasionally "0.0000" "0.0000"
## Frequently/Occasionally|Always/Almost always "0.0048" "0.0134"
```

```
model_knowledge <- multinom(
  diversity_lab ~ skin_type_lab + gender_lab + apply_freq_lab +
    summer_apply_freq_lab + incentive_lab,
  data = data_labeled, trace = FALSE
)

# Extract coef table pieces
sm <- summary(model_knowledge)
bet <- sm$coefficients # rows = outcomes (vs baseline), cols = predictors
se <- sm$standard.errors

# z and two-sided p-values
z <- bet / se
p <- 2 * (1 - pnorm(abs(z)))

# Build long-form table like your polr ctable
ctable <- data.frame(
  Outcome = rep(rownames(bet), each = ncol(bet)),
  Predictor = rep(colnames(bet), times = nrow(bet)),
```

```

Value      = as.vector(t(bet)),
`Std. Error` = as.vector(t(se)),
`z value`   = as.vector(t(z)),
check.names = FALSE
)

# Add raw and BH-adjusted p-values (across all coefficients in this model)
p_vec <- as.vector(t(p))
ctable$`p value` <- formatC(p_vec, format = "f", digits = 4)
ctable$`adj p (BH)` <- formatC(p.adjust(p_vec, method = "BH"), format = "f", digits = 4)

# Final column order
ctable <- ctable[, c("Outcome", "Predictor", "Value", "Std. Error", "z value", "p value", "adj p (BH)")]

ctable

```

##	Outcome	Predictor	Value	Std. Error
## 1	Incorrect	(Intercept)	0.607078966	2.188119e-01
## 2	Incorrect	skin_type_lab.L	0.372260811	3.955279e-01
## 3	Incorrect	skin_type_lab.Q	-0.696279306	3.654240e-01
## 4	Incorrect	skin_type_lab.C	-0.048313360	3.178669e-01
## 5	Incorrect	skin_type_lab^4	0.005891662	2.749171e-01
## 6	Incorrect	skin_type_lab^5	-0.032199932	2.490264e-01
## 7	Incorrect	gender_labFemale	0.532562295	2.283891e-01
## 8	Incorrect	gender_labGender expansive	14.529757984	9.667795e-08
## 9	Incorrect	gender_labNonbinary	0.988964742	8.459319e-01
## 10	Incorrect	gender_labOther	13.393364724	1.297482e-07
## 11	Incorrect	gender_labPrefer not to say	1.253197224	1.093935e+00
## 12	Incorrect	apply_freq_lab.L	-0.048247302	3.299803e-01
## 13	Incorrect	apply_freq_lab.Q	-0.341608388	3.141405e-01
## 14	Incorrect	apply_freq_lab.C	0.779540026	2.988452e-01
## 15	Incorrect	apply_freq_lab^4	-0.331953781	2.849302e-01
## 16	Incorrect	apply_freq_lab^5	0.112677955	2.927531e-01
## 17	Incorrect	summer_apply_freq_lab.L	0.284966586	2.933235e-01
## 18	Incorrect	summer_apply_freq_lab.Q	-0.620883679	2.904285e-01
## 19	Incorrect	summer_apply_freq_lab.C	0.401532221	2.895968e-01
## 20	Incorrect	incentive_labPrevent skin cancer	0.319956107	2.248294e-01
## 21	Incorrect	incentive_labPrevent wrinkles	0.500680984	3.574971e-01
## 22	Incorrect	incentive_labOther Incentive	14.145767432	5.616346e-01
## 23	Unsure	(Intercept)	-1.184324407	3.720490e-01
## 24	Unsure	skin_type_lab.L	0.654193612	7.785437e-01
## 25	Unsure	skin_type_lab.Q	-0.588865570	7.454815e-01
## 26	Unsure	skin_type_lab.C	-0.693368100	5.831398e-01
## 27	Unsure	skin_type_lab^4	-0.787019018	4.740602e-01
## 28	Unsure	skin_type_lab^5	0.777777577	4.910634e-01
## 29	Unsure	gender_labFemale	-0.166939098	3.743480e-01
## 30	Unsure	gender_labGender expansive	-2.842443178	4.495593e-09
## 31	Unsure	gender_labNonbinary	-14.484927500	2.042716e-07
## 32	Unsure	gender_labOther	-3.883736306	2.041466e-09
## 33	Unsure	gender_labPrefer not to say	1.026258315	1.497190e+00
## 34	Unsure	apply_freq_lab.L	-0.181483524	5.439593e-01
## 35	Unsure	apply_freq_lab.Q	-0.083599711	5.301873e-01
## 36	Unsure	apply_freq_lab.C	0.515842167	4.922022e-01

## 37	Unsure	apply_freq_lab^4	-0.095329775	4.613060e-01
## 38	Unsure	apply_freq_lab^5	-0.495955456	4.864649e-01
## 39	Unsure	summer_apply_freq_lab.L	0.108003004	5.441621e-01
## 40	Unsure	summer_apply_freq_lab.Q	-1.249719149	4.792914e-01
## 41	Unsure	summer_apply_freq_lab.C	0.462511014	4.184315e-01
## 42	Unsure	incentive_labPrevent skin cancer	-0.140603272	4.055621e-01
## 43	Unsure	incentive_labPrevent wrinkles	0.924376583	5.321383e-01
## 44	Unsure	incentive_labOther Incentive	14.518954491	5.616342e-01
##		z value	p value	adj p (BH)
## 1		2.774433e+00	0.0055	0.0270
## 2		9.411745e-01	0.3466	0.5084
## 3		-1.905401e+00	0.0567	0.1783
## 4		-1.519924e-01	0.8792	0.9180
## 5		2.143069e-02	0.9829	0.9829
## 6		-1.293033e-01	0.8971	0.9180
## 7		2.331820e+00	0.0197	0.0723
## 8		1.502903e+08	0.0000	0.0000
## 9		1.169083e+00	0.2424	0.4619
## 10		1.032258e+08	0.0000	0.0000
## 11		1.145586e+00	0.2520	0.4619
## 12		-1.462127e-01	0.8838	0.9180
## 13		-1.087438e+00	0.2768	0.4685
## 14		2.608508e+00	0.0091	0.0365
## 15		-1.165036e+00	0.2440	0.4619
## 16		3.848907e-01	0.7003	0.8784
## 17		9.715097e-01	0.3313	0.5027
## 18		-2.137819e+00	0.0325	0.1101
## 19		1.386522e+00	0.1656	0.3643
## 20		1.423106e+00	0.1547	0.3643
## 21		1.400518e+00	0.1614	0.3643
## 22		2.518678e+01	0.0000	0.0000
## 23		-3.183249e+00	0.0015	0.0080
## 24		8.402786e-01	0.4008	0.5688
## 25		-7.899130e-01	0.4296	0.5907
## 26		-1.189026e+00	0.2344	0.4619
## 27		-1.660167e+00	0.0969	0.2664
## 28		1.583864e+00	0.1132	0.2931
## 29		-4.459463e-01	0.6556	0.8485
## 30		-6.322732e+08	0.0000	0.0000
## 31		-7.091012e+07	0.0000	0.0000
## 32		-1.902425e+09	0.0000	0.0000
## 33		6.854563e-01	0.4931	0.6574
## 34		-3.336344e-01	0.7387	0.8784
## 35		-1.576796e-01	0.8747	0.9180
## 36		1.048029e+00	0.2946	0.4801
## 37		-2.066519e-01	0.8363	0.9180
## 38		-1.019509e+00	0.3080	0.4839
## 39		1.984758e-01	0.8427	0.9180
## 40		-2.607431e+00	0.0091	0.0365
## 41		1.105345e+00	0.2690	0.4685
## 42		-3.466874e-01	0.7288	0.8784
## 43		1.737098e+00	0.0824	0.2416
## 44		2.585127e+01	0.0000	0.0000

```

library(nnet)

# (Optional) set baseline for gender comparisons
# data_labeled$gender_lab <- relevel(data_labeled$gender_lab, ref = "Male")

# Fit multinomial model predicting gender
model_gender <- multinom(
  gender_lab ~ apply_freq_lab + incentive_lab + summer_apply_freq_lab + winter_apply_freq_lab + body_he
    price_lab + would_you_use_lab,
  data = data_labeled,
  trace = FALSE
)

# ----- Build output table (like your polr ctable) -----

sm <- summary(model_gender)
bet <- sm$coefficients          # rows = gender levels vs baseline; cols = predictors
se <- sm$standard.errors

# z and two-sided p-values
z <- bet / se
p <- 2 * (1 - pnorm(abs(z)))

# Long-form table
ctable <- data.frame(
  Outcome      = rep(rownames(bet), each = ncol(bet)),
  Predictor     = rep(colnames(bet), times = nrow(bet)),
  Value        = as.vector(t(bet)),
  `Std. Error` = as.vector(t(se)),
  `z value`    = as.vector(t(z)),
  check.names  = FALSE
)

# Raw + BH-adjusted p-values across all coefficients in this model
p_vec <- as.vector(t(p))
ctable$`p value` <- formatC(p_vec, format = "f", digits = 4)
ctable$`adj p (BH)` <- formatC(p.adjust(p_vec, method = "BH"), format = "f", digits = 4)

# Final column order
ctable <- cbind(ctable[, c("Outcome", "Predictor", "Value", "Std. Error", "z value")],
  ctable[, c("p value", "adj p (BH)")]
)

ctable

```

##	Outcome	Predictor	Value
## 1	Female	(Intercept)	0.68841410
## 2	Female	apply_freq_lab.L	-0.85713957
## 3	Female	apply_freq_lab.Q	-0.01339499
## 4	Female	apply_freq_lab.C	0.71541343
## 5	Female	apply_freq_lab^4	-0.10506305
## 6	Female	apply_freq_lab^5	0.12754023
## 7	Female	incentive_labPrevent skin cancer	0.08890674
## 8	Female	incentive_labPrevent wrinkles	0.50378286

## 9	Female	incentive_labOther Incentive	0.26045585
## 10	Female	summer_apply_freq_lab.L	0.89618141
## 11	Female	summer_apply_freq_lab.Q	-0.23018088
## 12	Female	summer_apply_freq_lab.C	-0.04222591
## 13	Female	winter_apply_freq_lab.L	0.17056012
## 14	Female	winter_apply_freq_lab.Q	-0.20024655
## 15	Female	winter_apply_freq_lab.C	-0.47699578
## 16	Female	body_headneck	-0.46665621
## 17	Female	body_torso	0.17155714
## 18	Female	body_limbs	-0.08372488
## 19	Female	price_labYes	0.20106004
## 20	Female	would_you_use_lab.L	-0.27676198
## 21	Female	would_you_use_lab.Q	-0.20467723
## 22	Gender expansive	(Intercept)	-71.36794769
## 23	Gender expansive	apply_freq_lab.L	-20.40338299
## 24	Gender expansive	apply_freq_lab.Q	22.92506754
## 25	Gender expansive	apply_freq_lab.C	-14.45508829
## 26	Gender expansive	apply_freq_lab^4	-42.37958952
## 27	Gender expansive	apply_freq_lab^5	0.31055683
## 28	Gender expansive	incentive_labPrevent skin cancer	-0.96534589
## 29	Gender expansive	incentive_labPrevent wrinkles	0.70830296
## 30	Gender expansive	incentive_labOther Incentive	-17.32603814
## 31	Gender expansive	summer_apply_freq_lab.L	20.35954840
## 32	Gender expansive	summer_apply_freq_lab.Q	-15.94686796
## 33	Gender expansive	summer_apply_freq_lab.C	6.19789078
## 34	Gender expansive	winter_apply_freq_lab.L	-22.29955306
## 35	Gender expansive	winter_apply_freq_lab.Q	-5.89075822
## 36	Gender expansive	winter_apply_freq_lab.C	-33.30277465
## 37	Gender expansive	body_headneck	20.93889884
## 38	Gender expansive	body_torso	-1.49142912
## 39	Gender expansive	body_limbs	0.07715273
## 40	Gender expansive	price_labYes	-0.75878294
## 41	Gender expansive	would_you_use_lab.L	-0.56226423
## 42	Gender expansive	would_you_use_lab.Q	0.03195896
## 43	Nonbinary	(Intercept)	-70.01399423
## 44	Nonbinary	apply_freq_lab.L	38.09450413
## 45	Nonbinary	apply_freq_lab.Q	-32.24297526
## 46	Nonbinary	apply_freq_lab.C	21.92079621
## 47	Nonbinary	apply_freq_lab^4	-11.62445063
## 48	Nonbinary	apply_freq_lab^5	4.52708297
## 49	Nonbinary	incentive_labPrevent skin cancer	0.41247260
## 50	Nonbinary	incentive_labPrevent wrinkles	0.81918443
## 51	Nonbinary	incentive_labOther Incentive	-18.75613688
## 52	Nonbinary	summer_apply_freq_lab.L	0.70242867
## 53	Nonbinary	summer_apply_freq_lab.Q	1.17065465
## 54	Nonbinary	summer_apply_freq_lab.C	-0.21272671
## 55	Nonbinary	winter_apply_freq_lab.L	-0.58701228
## 56	Nonbinary	winter_apply_freq_lab.Q	-1.37438616
## 57	Nonbinary	winter_apply_freq_lab.C	0.06499024
## 58	Nonbinary	body_headneck	54.89320723
## 59	Nonbinary	body_torso	1.48384269
## 60	Nonbinary	body_limbs	0.76281433
## 61	Nonbinary	price_labYes	1.66452333
## 62	Nonbinary	would_you_use_lab.L	-1.32925802

## 63	Nonbinary	would_you_use_lab.Q	-0.45210363	
## 64	Other	(Intercept)	-40.26123637	
## 65	Other	apply_freq_lab.L	-28.88532292	
## 66	Other	apply_freq_lab.Q	8.41734559	
## 67	Other	apply_freq_lab.C	13.35410379	
## 68	Other	apply_freq_lab^4	0.03543314	
## 69	Other	apply_freq_lab^5	-4.99602040	
## 70	Other	incentive_labPrevent skin cancer	-8.10558890	
## 71	Other	incentive_labPrevent wrinkles	-6.21008031	
## 72	Other	incentive_labOther Incentive	-7.88268510	
## 73	Other	summer_apply_freq_lab.L	-3.12350802	
## 74	Other	summer_apply_freq_lab.Q	-14.29870058	
## 75	Other	summer_apply_freq_lab.C	12.52021480	
## 76	Other	winter_apply_freq_lab.L	8.54279059	
## 77	Other	winter_apply_freq_lab.Q	-2.83709132	
## 78	Other	winter_apply_freq_lab.C	-9.41991159	
## 79	Other	body_headneck	-8.20242871	
## 80	Other	body_torso	12.90114770	
## 81	Other	body_limbs	-0.18050741	
## 82	Other	price_labYes	-7.71905111	
## 83	Other	would_you_use_lab.L	1.10503929	
## 84	Other	would_you_use_lab.Q	-9.34482489	
## 85	Prefer not to say	(Intercept)	-67.19192018	
## 86	Prefer not to say	apply_freq_lab.L	-27.78751381	
## 87	Prefer not to say	apply_freq_lab.Q	-25.62054023	
## 88	Prefer not to say	apply_freq_lab.C	-17.12783507	
## 89	Prefer not to say	apply_freq_lab^4	-7.99271583	
## 90	Prefer not to say	apply_freq_lab^5	-2.19894845	
## 91	Prefer not to say	incentive_labPrevent skin cancer	0.74664747	
## 92	Prefer not to say	incentive_labPrevent wrinkles	0.71104500	
## 93	Prefer not to say	incentive_labOther Incentive	-14.69614865	
## 94	Prefer not to say	summer_apply_freq_lab.L	-0.63313777	
## 95	Prefer not to say	summer_apply_freq_lab.Q	-0.07706098	
## 96	Prefer not to say	summer_apply_freq_lab.C	0.75623434	
## 97	Prefer not to say	winter_apply_freq_lab.L	12.82478438	
## 98	Prefer not to say	winter_apply_freq_lab.Q	27.84337252	
## 99	Prefer not to say	winter_apply_freq_lab.C	-38.61310548	
## 100	Prefer not to say	body_headneck	41.58434206	
## 101	Prefer not to say	body_torso	-1.95366668	
## 102	Prefer not to say	body_limbs	0.88665253	
## 103	Prefer not to say	price_labYes	1.56775069	
## 104	Prefer not to say	would_you_use_lab.L	-0.78035065	
## 105	Prefer not to say	would_you_use_lab.Q	-0.32152395	
##	Std. Error	z value	p value adj p (BH)	
## 1	5.895210e-01	1.167752e+00	0.2429	0.4637
## 2	2.917786e-01	-2.937637e+00	0.0033	0.0089
## 3	2.790455e-01	-4.800288e-02	0.9617	0.9916
## 4	2.632203e-01	2.717926e+00	0.0066	0.0168
## 5	2.479726e-01	-4.236882e-01	0.6718	0.8499
## 6	2.568256e-01	4.966025e-01	0.6195	0.8030
## 7	1.980376e-01	4.489387e-01	0.6535	0.8368
## 8	3.138455e-01	1.605194e+00	0.1085	0.2372
## 9	8.614616e-01	3.023418e-01	0.7624	0.9418
## 10	2.858622e-01	3.135012e+00	0.0017	0.0047

## 11	2.612431e-01	-8.810985e-01	0.3783	0.6383
## 12	2.363647e-01	-1.786473e-01	0.8582	0.9711
## 13	2.108535e-01	8.089036e-01	0.4186	0.6383
## 14	2.355119e-01	-8.502607e-01	0.3952	0.6383
## 15	2.887328e-01	-1.652032e+00	0.0985	0.2201
## 16	5.567426e-01	-8.381902e-01	0.4019	0.6383
## 17	2.149562e-01	7.981027e-01	0.4248	0.6383
## 18	2.155963e-01	-3.883409e-01	0.6978	0.8722
## 19	1.911669e-01	1.051751e+00	0.2929	0.5344
## 20	2.128378e-01	-1.300342e+00	0.1935	0.3983
## 21	1.990256e-01	-1.028396e+00	0.3038	0.5406
## 22	3.656125e-01	-1.952011e+02	0.0000	0.0000
## 23	6.015660e-01	-3.391711e+01	0.0000	0.0000
## 24	4.240394e-01	5.406354e+01	0.0000	0.0000
## 25	7.121973e-01	-2.029647e+01	0.0000	0.0000
## 26	5.090426e-01	-8.325353e+01	0.0000	0.0000
## 27	4.057202e-01	7.654459e-01	0.4440	0.6483
## 28	1.179573e+00	-8.183856e-01	0.4131	0.6383
## 29	1.388964e+00	5.099506e-01	0.6101	0.8007
## 30	6.383807e-09	-2.714061e+09	0.0000	0.0000
## 31	7.456932e-01	2.730285e+01	0.0000	0.0000
## 32	8.726291e-01	-1.827451e+01	0.0000	0.0000
## 33	8.109168e-01	7.643066e+00	0.0000	0.0000
## 34	5.270679e-01	-4.230869e+01	0.0000	0.0000
## 35	5.920930e-01	-9.949041e+00	0.0000	0.0000
## 36	3.258707e-01	-1.021963e+02	0.0000	0.0000
## 37	3.656125e-01	5.727074e+01	0.0000	0.0000
## 38	1.257663e+00	-1.185873e+00	0.2357	0.4637
## 39	1.016328e+00	7.591321e-02	0.9395	0.9916
## 40	9.522749e-01	-7.968108e-01	0.4256	0.6383
## 41	9.314771e-01	-6.036265e-01	0.5461	0.7545
## 42	8.240590e-01	3.878237e-02	0.9691	0.9916
## 43	5.434191e-01	-1.288398e+02	0.0000	0.0000
## 44	6.937978e-01	5.490722e+01	0.0000	0.0000
## 45	7.595312e-01	-4.245115e+01	0.0000	0.0000
## 46	7.614891e-01	2.878675e+01	0.0000	0.0000
## 47	7.681721e-01	-1.513261e+01	0.0000	0.0000
## 48	9.289810e-01	4.873170e+00	0.0000	0.0000
## 49	7.343517e-01	5.616827e-01	0.5743	0.7731
## 50	9.562407e-01	8.566718e-01	0.3916	0.6383
## 51	1.423159e-09	-1.317922e+10	0.0000	0.0000
## 52	8.632109e-01	8.137393e-01	0.4158	0.6383
## 53	8.242430e-01	1.420279e+00	0.1555	0.3266
## 54	7.851800e-01	-2.709273e-01	0.7864	0.9492
## 55	9.607672e-01	-6.109829e-01	0.5412	0.7545
## 56	8.821426e-01	-1.558009e+00	0.1192	0.2555
## 57	9.098429e-01	7.143018e-02	0.9431	0.9916
## 58	5.434191e-01	1.010145e+02	0.0000	0.0000
## 59	7.466176e-01	1.987420e+00	0.0469	0.1145
## 60	8.244335e-01	9.252588e-01	0.3548	0.6210
## 61	7.653757e-01	2.174779e+00	0.0296	0.0741
## 62	6.750582e-01	-1.969101e+00	0.0489	0.1168
## 63	5.913562e-01	-7.645200e-01	0.4446	0.6483
## 64	4.526659e+01	-8.894250e-01	0.3738	0.6383

```
## 65 1.807677e+00 -1.597925e+01 0.0000 0.0000
## 66 4.444773e+01 1.893763e-01 0.8498 0.9711
## 67 7.759676e+01 1.720962e-01 0.8634 0.9711
## 68 7.128420e+01 4.970687e-04 0.9996 0.9996
## 69 3.706783e+01 -1.347805e-01 0.8928 0.9779
## 70 1.442953e-03 -5.617364e+03 0.0000 0.0000
## 71 1.704290e-04 -3.643792e+04 0.0000 0.0000
## 72 2.704217e-04 -2.914961e+04 0.0000 0.0000
## 73 8.434550e+01 -3.703230e-02 0.9705 0.9916
## 74 8.298252e+01 -1.723098e-01 0.8632 0.9711
## 75 1.687328e+01 7.420144e-01 0.4581 0.6589
## 76 6.413647e+01 1.331971e-01 0.8940 0.9779
## 77 8.290245e+01 -3.422204e-02 0.9727 0.9916
## 78 5.728250e+01 -1.644466e-01 0.8694 0.9711
## 79 4.526607e+01 -1.812048e-01 0.8562 0.9711
## 80 4.526362e+01 2.850224e-01 0.7756 0.9470
## 81 4.526475e+01 -3.987814e-03 0.9968 0.9996
## 82 1.489208e-03 -5.183325e+03 0.0000 0.0000
## 83 4.632750e-03 2.385277e+02 0.0000 0.0000
## 84 3.695087e+01 -2.528987e-01 0.8003 0.9550
## 85 4.948543e-01 -1.357812e+02 0.0000 0.0000
## 86 7.483239e-01 -3.713300e+01 0.0000 0.0000
## 87 6.397477e-01 -4.004788e+01 0.0000 0.0000
## 88 8.308632e-01 -2.061451e+01 0.0000 0.0000
## 89 8.227374e-01 -9.714784e+00 0.0000 0.0000
## 90 7.525822e-01 -2.921872e+00 0.0035 0.0091
## 91 7.132500e-01 1.046824e+00 0.2952 0.5344
## 92 1.228166e+00 5.789485e-01 0.5626 0.7672
## 93 5.324435e-08 -2.760133e+08 0.0000 0.0000
## 94 9.588503e-01 -6.603093e-01 0.5091 0.7223
## 95 8.796540e-01 -8.760374e-02 0.9302 0.9916
## 96 7.075725e-01 1.068773e+00 0.2852 0.5344
## 97 7.382793e-01 1.737118e+01 0.0000 0.0000
## 98 6.063856e-01 4.591694e+01 0.0000 0.0000
## 99 5.303398e-01 -7.280824e+01 0.0000 0.0000
## 100 4.948543e-01 8.403350e+01 0.0000 0.0000
## 101 1.099910e+00 -1.776205e+00 0.0757 0.1728
## 102 7.558501e-01 1.173053e+00 0.2408 0.4637
## 103 8.346467e-01 1.878341e+00 0.0603 0.1408
## 104 6.585428e-01 -1.184966e+00 0.2360 0.4637
## 105 6.125820e-01 -5.248668e-01 0.5997 0.7970
```

```
library(nnet)
```

```
# Optional: Set baseline for incentive comparisons (e.g., "Prevent sunburn")
# data_labeled$incentive_lab <- relevel(data_labeled$incentive_lab, ref = "Prevent sunburn")

# Fit multinomial logistic regression model
model_incentive_small <- multinom(
  summer_apply_freq_lab ~ incentive_lab + price_lab + gender_lab,
  data = data_labeled,
  trace = FALSE
)
```

```

# ----- Build clean coefficient table -----

sm <- summary(model_incentive_small)
bet <- sm$coefficients
se <- sm$standard.errors

# Calculate z-values and p-values
z <- bet / se
p <- 2 * (1 - pnorm(abs(z)))

# Build long-form table
ctable <- data.frame(
  Outcome      = rep(rownames(bet), each = ncol(bet)),
  Predictor     = rep(colnames(bet), times = nrow(bet)),
  Value        = as.vector(t(bet)),
  `Std. Error` = as.vector(t(se)),
  `z value`    = as.vector(t(z)),
  check.names  = FALSE
)

# Add raw & BH-adjusted p-values
p_vec <- as.vector(t(p))
ctable$`p value` <- formatC(p_vec, format = "f", digits = 4)
ctable$`adj p (BH)` <- formatC(p.adjust(p_vec, method = "BH"), format = "f", digits = 4)

# Final clean table formatting
ctable <- cbind(ctable[, c("Outcome", "Predictor", "Value", "Std. Error", "z value")],
               ctable[, c("p value", "adj p (BH)")]

ctable

```

	Outcome	Predictor	Value
## 1	Specific outdoor occasions	(Intercept)	0.21297314
## 2	Specific outdoor occasions	incentive_labPrevent skin cancer	0.03072943
## 3	Specific outdoor occasions	incentive_labPrevent wrinkles	0.13933004
## 4	Specific outdoor occasions	incentive_labOther Incentive	-17.42536885
## 5	Specific outdoor occasions	price_labYes	-0.19234192
## 6	Specific outdoor occasions	gender_labFemale	0.39721590
## 7	Specific outdoor occasions	gender_labGender expansive	12.05232238
## 8	Specific outdoor occasions	gender_labNonbinary	-0.09387997
## 9	Specific outdoor occasions	gender_labOther	16.13016902
## 10	Specific outdoor occasions	gender_labPrefer not to say	1.01073612
## 11	Frequently/Occasionally	(Intercept)	0.43584973
## 12	Frequently/Occasionally	incentive_labPrevent skin cancer	0.11487354
## 13	Frequently/Occasionally	incentive_labPrevent wrinkles	0.63687163
## 14	Frequently/Occasionally	incentive_labOther Incentive	-0.18752432
## 15	Frequently/Occasionally	price_labYes	0.25512514
## 16	Frequently/Occasionally	gender_labFemale	1.25149396
## 17	Frequently/Occasionally	gender_labGender expansive	13.02350930
## 18	Frequently/Occasionally	gender_labNonbinary	-0.34489469
## 19	Frequently/Occasionally	gender_labOther	-4.55207778
## 20	Frequently/Occasionally	gender_labPrefer not to say	0.34749108
## 21	Always/Almost always	(Intercept)	0.74219473
## 22	Always/Almost always	incentive_labPrevent skin cancer	0.30849393

## 23	Always/Almost always	incentive_lab	Prevent wrinkles	1.15749764	
## 24	Always/Almost always	incentive_lab	Other Incentive	-0.51096663	
## 25	Always/Almost always		price_lab	Yes	0.24259868
## 26	Always/Almost always		gender_lab	Female	1.64372212
## 27	Always/Almost always		gender_lab	Gender expansive	11.19434033
## 28	Always/Almost always		gender_lab	Nonbinary	-0.11033987
## 29	Always/Almost always		gender_lab	Other	-5.19667549
## 30	Always/Almost always		gender_lab	Prefer not to say	0.17624891
##	Std. Error	z value	p value	adj p (BH)	
## 1	3.421833e-01	6.223950e-01	0.5337	0.8427	
## 2	3.945303e-01	7.788866e-02	0.9379	0.9379	
## 3	7.771476e-01	1.792839e-01	0.8577	0.9379	
## 4	9.536998e-08	-1.827133e+08	0.0000	0.0000	
## 5	3.788621e-01	-5.076832e-01	0.6117	0.9175	
## 6	3.949812e-01	1.005658e+00	0.3146	0.6954	
## 7	7.850728e-01	1.535185e+01	0.0000	0.0000	
## 8	1.039823e+00	-9.028459e-02	0.9281	0.9379	
## 9	1.841858e-08	8.757554e+08	0.0000	0.0000	
## 10	1.191057e+00	8.486044e-01	0.3961	0.7317	
## 11	3.114825e-01	1.399275e+00	0.1617	0.4043	
## 12	3.441797e-01	3.337604e-01	0.7386	0.9379	
## 13	6.605481e-01	9.641563e-01	0.3350	0.6954	
## 14	1.201725e+00	-1.560460e-01	0.8760	0.9379	
## 15	3.279591e-01	7.779176e-01	0.4366	0.7317	
## 16	3.431842e-01	3.646712e+00	0.0003	0.0009	
## 17	5.900781e-01	2.207082e+01	0.0000	0.0000	
## 18	9.484251e-01	-3.636499e-01	0.7161	0.9379	
## 19	8.398891e-11	-5.419856e+10	0.0000	0.0000	
## 20	1.184373e+00	2.933965e-01	0.7692	0.9379	
## 21	2.960345e-01	2.507122e+00	0.0122	0.0365	
## 22	3.285290e-01	9.390159e-01	0.3477	0.6954	
## 23	6.355240e-01	1.821328e+00	0.0686	0.1870	
## 24	1.176771e+00	-4.342108e-01	0.6641	0.9379	
## 25	3.134777e-01	7.738945e-01	0.4390	0.7317	
## 26	3.274852e-01	5.019226e+00	0.0000	0.0000	
## 27	7.780185e-01	1.438827e+01	0.0000	0.0000	
## 28	8.562174e-01	-1.288690e-01	0.8975	0.9379	
## 29	5.760510e-11	-9.021207e+10	0.0000	0.0000	
## 30	1.148487e+00	1.534618e-01	0.8780	0.9379	

```

# Fit ordinal logistic regression model
model_apply <- polr(apply_freq_lab ~ school_lab + gender_lab + seen_dispenser_lab +
                      would_you_use_lab + aid_in_prevention_lab +
                      skin_type_lab + price_lab + incentive_lab +
                      diversity_lab +
                      winter_apply_freq_lab + spring_apply_freq_lab +
                      summer_apply_freq_lab + fall_apply_freq_lab,
                    data = data_labeled, Hess = TRUE)

# View coefficient estimates and z-tests
summary(model_apply)
# Convert to p-values
# Coefficients with t-values
(ctable <- coef(summary(model_apply)))

```

```

# Compute 2-sided p-values
p_values <- 2 * pnorm(abs(ctable[, "t value"]), lower.tail = FALSE)

# Add them to table
ctable <- cbind(ctable, "p value" = p_values)

# Format p-values in decimal form with up to 4 decimals (adjust as you like)
ctable[, "p value"] <- formatC(p_values, format = "f", digits = 4)

print(ctable)

# 1. Ordered coding (default, polynomial contrasts)
model_ordered <- polr(apply_freq_lab ~ skin_type_lab,
                      data = data_labeled, Hess = TRUE)
summary(model_ordered)

# 2. Nominal coding (dummy contrasts)
data_labeled$skin_type_nom <- factor(data_labeled$skin_type_lab, ordered = FALSE)

model_nominal <- polr(apply_freq_lab ~ skin_type_nom,
                     data = data_labeled, Hess = TRUE)

# Fit the Proportional Odds Model
model_1 <- polr(would_you_use_ord ~ seen_free_dispenser_lab + aid_in_prevention_lab + price_lab,
               data = df_clean, Hess = TRUE) # Hess=TRUE allows us to get p-values later

# View a summary of the model
summary(model_1)

summary(model_ordered)
summary(model_nominal)

# force all to nominal (unordered factors)
data_labeled$school_nom <- factor(data_labeled$school_lab, ordered = FALSE)
data_labeled$gender_nom <- factor(data_labeled$gender_lab, ordered = FALSE)
data_labeled$seen_dispenser_nom <- factor(data_labeled$seen_dispenser_lab, ordered = FALSE)
data_labeled$would_use_nom <- factor(data_labeled$would_you_use_lab, ordered = FALSE)
data_labeled$aid_nom <- factor(data_labeled$aid_in_prevention_lab, ordered = FALSE)
data_labeled$skin_type_nom <- factor(data_labeled$skin_type_lab, ordered = FALSE)
data_labeled$price_nom <- factor(data_labeled$price_lab, ordered = FALSE)
data_labeled$incentive_nom <- factor(data_labeled$incentive_lab, ordered = FALSE)
data_labeled$diversity_nom <- factor(data_labeled$diversity_lab, ordered = FALSE)
data_labeled$winter_nom <- factor(data_labeled$winter_apply_freq_lab, ordered = FALSE)
data_labeled$spring_nom <- factor(data_labeled$spring_apply_freq_lab, ordered = FALSE)
data_labeled$summer_nom <- factor(data_labeled$summer_apply_freq_lab, ordered = FALSE)
data_labeled$fall_nom <- factor(data_labeled$fall_apply_freq_lab, ordered = FALSE)

model_nom <- polr(apply_freq_lab ~ school_nom + gender_nom + seen_dispenser_nom +
                 would_use_nom + aid_nom + skin_type_nom +
                 price_nom + incentive_nom + diversity_nom +
                 winter_nom + spring_nom + summer_nom + fall_nom,

```

```

        data = data_labeled, Hess = TRUE)

summary(model_nom)

# Extract coefficients
(ctable <- coef(summary(model_nom)))

# Raw 2-sided p-values
p_values <- 2 * pnorm(abs(ctable[, "t value"]), lower.tail = FALSE)

# Adjust them (choose method: "BH" = FDR, "holm" = step-down, "bonferroni" = strictest)
p_values_adj <- p.adjust(p_values, method = "BH")

# Bind both to the coefficient table
ctable <- cbind(ctable,
  "p value" = formatC(p_values, format = "f", digits = 4),
  "adj p (BH)" = formatC(p_values_adj, format = "f", digits = 4))

print(ctable)

# quick one-way summaries
library(janitor); library(dplyr)
tab1 <- data_labeled %>% tabyl(would_you_use_lab) %>% adorn_totals("row")
tab2 <- data_labeled %>% tabyl(aid_in_prevention_lab) %>% adorn_totals("row")
tab3 <- data_labeled %>% tabyl(seen_dispenser_lab) %>% adorn_totals("row")
tab4 <- data_labeled %>% tabyl(price_lab) %>% adorn_totals("row")
tab_skin <- data_labeled %>% tabyl(skin_type_lab) %>% adorn_totals("row")
tab1; tab2; tab3; tab4; tab_skin

# bar charts
library(ggplot2)

p_use <- ggplot(data_labeled, aes(would_you_use_lab)) +
  geom_bar() + labs(title="Would You Use Free Dispenser?", x="Response", y="Count")

p_aid <- ggplot(data_labeled, aes(aid_in_prevention_lab)) +
  geom_bar() + labs(title="Does Free Dispenser Aid Prevention?", x="Response", y="Count")

p_skin <- ggplot(data_labeled, aes(skin_type_lab)) +
  geom_bar() + coord_flip() +
  labs(title="Skin Type Distribution", x="", y="Count")

dir.create("outputs", showWarnings = FALSE)
ggsave("outputs/would_you_use_bar.png", p_use, width=6, height=4, dpi=300)
ggsave("outputs/aid_prevention_bar.png", p_aid, width=6, height=4, dpi=300)
ggsave("outputs/skin_type_bar.png", p_skin, width=6, height=4, dpi=300)

p_use; p_aid; p_skin

# associations (chi-square + ordinal trend checks)

# 1) Seen dispenser (binary) vs Would use (ordinal)

```

```

tab_seen_use <- table(data_labeled$seen_dispenser_lab, data_labeled$would_you_use_lab)
chi_seen <- chisq.test(tab_seen_use)

# 2) Skin type (ordinal) vs Would use (ordinal)
tab_skin_use <- table(data_labeled$skin_type_lab, data_labeled$would_you_use_lab)
chi_skin <- chisq.test(tab_skin_use)

# ordinal-to-ordinal monotonic association (Spearman)
spearman_skin_use <- cor(as.numeric(data_labeled$skin_type_lab),
                        as.numeric(data_labeled$would_you_use_lab),
                        use="complete.obs", method="spearman")

# 3) Aid in prevention vs Would use (ordinal)
tab_aid_use <- table(data_labeled$aid_in_prevention_lab, data_labeled$would_you_use_lab)
chi_aid <- chisq.test(tab_aid_use)

spearman_aid_use <- cor(as.numeric(data_labeled$aid_in_prevention_lab),
                      as.numeric(data_labeled$would_you_use_lab),
                      use="complete.obs", method="spearman")

# Collect chi-square p-values
raw_p <- c(chi_seen$p.value, chi_skin$p.value, chi_aid$p.value)
names(raw_p) <- c("Seen vs Use", "Skin vs Use", "Aid vs Use")

# Bonferroni adjustment
pvals_bonf <- p.adjust(raw_p, method = "bonferroni")

# Combine into a summary table
results <- data.frame(
  Test = names(raw_p),
  Raw_p = raw_p,
  Bonferroni_p = pvals_bonf,
  row.names = NULL
)

print(results)

# Optional: also view Spearman results
spearman_results <- data.frame(
  Variable = c("Skin vs Use", "Aid vs Use"),
  Spearman_rho = c(spearman_skin_use, spearman_aid_use)
)

print(spearman_results)

library(MASS)

tab3 <- xtabs(~ seen_dispenser_lab + aid_in_prevention_lab + would_you_use_lab,
             data = data_labeled)

# Fit saturated log-linear model (all interactions)
mod_full <- loglm(~ seen_dispenser_lab * aid_in_prevention_lab * would_you_use_lab,
                 data = tab3)

```

```

# Drop the 3-way interaction and compare
mod_no3 <- loglm(~ (seen_dispenser_lab + aid_in_prevention_lab + would_you_use_lab)^2,
                 data = tab3)

anova(mod_no3, mod_full) # tests significance of the 3-way interaction

# Libraries
library(vcd)
library(reshape2)
library(ggplot2)

# Select categorical variables only
cat_vars <- data_labeled[, c("would_you_use_lab", "aid_in_prevention_lab",
                             "seen_dispenser_lab", "price_lab",
                             "knowledge_lab", "gender_lab",
                             "school_lab", "incentive_lab",
                             "skin_type_lab", "apply_freq_lab",
                             "winter_apply_freq_lab", "spring_apply_freq_lab",
                             "summer_apply_freq_lab", "fall_apply_freq_lab")]

# Create an empty matrix for Cramér's V
n <- length(cat_vars)
cramer_matrix <- matrix(NA, nrow = n, ncol = n,
                        dimnames = list(names(cat_vars), names(cat_vars)))

# Loop through all pairs of variables
for (i in 1:n) {
  for (j in 1:n) {
    if (i != j) {
      tbl <- table(cat_vars[[i]], cat_vars[[j]])
      chi <- suppressWarnings(chisq.test(tbl))
      # Calculate Cramér's V = sqrt(chi^2 / (n * (min(r,c)-1)))
      v <- sqrt(chi$statistic / (sum(tbl) * (min(nrow(tbl), ncol(tbl)) - 1)))
      cramer_matrix[i, j] <- as.numeric(v)
    } else {
      cramer_matrix[i, j] <- 1 # Perfect association with itself
    }
  }
}

# Convert to data frame for ggplot
cramer_df <- melt(cramer_matrix, varnames = c("Var1", "Var2"), value.name = "CramersV")

ggplot(cramer_df, aes(x = Var1, y = Var2, fill = CramersV)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(low = "#17497b", mid = "#F7F7F7", high = "#b21880",
                      midpoint = 0.3, limit = c(0,1), name = "Cramér's V") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        axis.title.x = element_blank(),
        axis.title.y = element_blank()) +
  ggtitle("Cramér's V Association Matrix for Categorical Variables")

```

```

# Libraries
library(vcd)
library(reshape2)
library(ggplot2)

# Select categorical variables only
cat_vars <- data_labeled[, c("would_you_use_lab", "aid_in_prevention_lab",
                             "seen_dispenser_lab", "price_lab",
                             "knowledge_lab", "gender_lab",
                             "school_lab", "incentive_lab",
                             "skin_type_lab"), "apply"]

# Create an empty matrix for Cramér's V
n <- length(cat_vars)
cramer_matrix <- matrix(NA, nrow = n, ncol = n,
                        dimnames = list(names(cat_vars), names(cat_vars)))

# Loop through all pairs of variables
for (i in 1:n) {
  for (j in 1:n) {
    if (i != j) {
      tbl <- table(cat_vars[[i]], cat_vars[[j]])
      chi <- suppressWarnings(chisq.test(tbl))
      # Calculate Cramér's V = sqrt(chi^2 / (n * (min(r,c)-1)))
      v <- sqrt(chi$statistic / (sum(tbl) * (min(nrow(tbl), ncol(tbl)) - 1)))
      cramer_matrix[i, j] <- as.numeric(v)
    } else {
      cramer_matrix[i, j] <- 1 # Perfect association with itself
    }
  }
}

# Convert to data frame for ggplot
cramer_df <- melt(cramer_matrix, varnames = c("Var1", "Var2"), value.name = "CramersV")

ggplot(cramer_df, aes(x = Var1, y = Var2, fill = CramersV)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(low = "#17497b", mid = "#F7F7F7", high = "#b21880",
                      midpoint = 0.3, limit = c(0,1), name = "Cramér's V") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        axis.title.x = element_blank(),
        axis.title.y = element_blank()) +
  ggtitle("Cramér's V Association Matrix for Categorical Variables")

## Add the three multiselcect multinomail variables into the mix
where_labels <- label_where_dispenser()
colnames(dat_labeled)[grep("^where_", colnames(dat_labeled))] <- where_labels

body_labels <- label_body_parts()
colnames(dat_labeled)[grep("^body_part_", colnames(dat_labeled))] <- body_labels

reason_labels <- label_reasons_against()

```

```
colnames(dat_labeled)[grep("^reasons_against_", colnames(dat_labeled))] <- reason_labels
```

```
library(dplyr)
library(vcd)      # assocstats()
library(reshape2) # melt()
library(ggplot2)

# --- 1) Base labeled categorical variables (from your earlier matrix) -----
base_vars <- c(
  "would_you_use_lab", "aid_in_prevention_lab",
  "seen_dispenser_lab", "price_lab",
  "knowledge_lab", "gender_lab",
  "school_lab", "incentive_lab",
  "skin_type_lab"
)

# --- 2) Add multi-select binaries, excluding "Other" for where/reasons -----

# Use your label helpers to get the intended column names
where_keep <- setdiff(label_where_dispenser(), "Other")
body_keep  <- label_body_parts()           # no "Other" here
reason_keep <- setdiff(label_reasons_against(), "Other")

# Intersect with actual columns (you already renamed these to labels earlier)
existing <- names(dat_labeled)
where_cols <- intersect(where_keep, existing)
body_cols <- intersect(body_keep, existing)
reason_cols <- intersect(reason_keep, existing)

multi_cols <- c(where_cols, body_cols, reason_cols)

# --- 3) Build the full analysis frame -----
all_vars <- c(base_vars, multi_cols)
cat_df <- dat_labeled[, all_vars, drop = FALSE]

# Coerce everything to factors (0/1 binaries become two-level factors)
for (nm in names(cat_df)) {
  if (!is.factor(cat_df[[nm]])) cat_df[[nm]] <- factor(cat_df[[nm]])
}

# --- 4) Robust Cramér's V function -----
cramers_v <- function(x, y) {
  ok <- complete.cases(x, y)
  if (!any(ok)) return(NA_real_)
  x <- droplevels(x[ok]); y <- droplevels(y[ok])
  if (nlevels(x) < 2 || nlevels(y) < 2) return(NA_real_)
  tbl <- table(x, y)
  if (any(dim(tbl) < 2)) return(NA_real_)
  vcd::assocstats(tbl)$cramer
}

# --- 5) Compute the big matrix -----
p <- ncol(cat_df)
```

```

V <- matrix(NA_real_, p, p, dimnames = list(names(cat_df), names(cat_df)))

for (i in seq_len(p)) {
  for (j in seq_len(p)) {
    V[i, j] <- if (i == j) 1 else cramers_v(cat_df[[i]], cat_df[[j]])
  }
}

# --- 6) Heatmap -----
V_melt <- melt(V, varnames = c("Var1", "Var2"), value.name = "CramersV")

ggplot(V_melt, aes(Var1, Var2, fill = CramersV)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(low = "#17497b", mid = "#F7F7F7", high = "#b21880",
                      midpoint = 0.30, limits = c(0,1),
                      name = "Cramér's V", na.value = "grey90") +
  labs(title = "Cramér's V - All categorical + multi-select binaries (no 'Other')",
       x = NULL, y = NULL) +
  theme_minimal(base_size = 12) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        panel.grid = element_blank())

# --- 7) Top pairs to inspect next -----
V_melt %>%
  filter(!is.na(CramersV), Var1 < Var2) %>%
  arrange(desc(CramersV)) %>%
  slice_head(n = 25)

```

```

library(grid) # for unit()

# Bigger, clearer heatmap (no reordering)
p <- ggplot(V_melt, aes(Var1, Var2, fill = CramersV)) +
  geom_tile(color = "white", linewidth = 0.2) +
  scale_fill_gradient2(
    low = "#17497b", mid = "#F7F7F7", high = "#b21880",
    midpoint = 0.30, limits = c(0, 1), name = "Cramér's V",
    na.value = "grey90"
  ) +
  labs(
    title = "Cramér's V - All categorical + multi-select binaries (no 'Other')",
    x = NULL, y = NULL
  ) +
  theme_minimal(base_size = 16) + # bump global text size
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1, vjust = 1, size = 11),
    axis.text.y = element_text(size = 11),
    plot.title = element_text(size = 20, face = "bold"),
    panel.grid = element_blank(),
    plot.margin = unit(c(10, 30, 10, 10), "mm")
  ) +
  coord_fixed() # square tiles so colors aren't distorted

```

p

```
ggsave("cramersV_heatmap_all.png", p, width = 20, height = 18, dpi = 300)
```

```
library(ggplot2)
library(reshape2)
library(dplyr)

### 1. Define groups based on labels you used earlier
body_keep <- label_body_parts()
where_keep <- setdiff(label_where_dispenser(), "Other Location")
reason_keep <- setdiff(label_reasons_against(), "Other Reason")

### 2. Filter the big Cramér's V matrix for each group
V_body <- V[rownames(V) %in% body_keep, colnames(V) %in% body_keep]
V_where <- V[rownames(V) %in% where_keep, colnames(V) %in% where_keep]
V_reason <- V[rownames(V) %in% reason_keep, colnames(V) %in% reason_keep]

### 3. Function to plot a heatmap given a sub-matrix
plot_heatmap <- function(V_sub, title) {
  df <- melt(V_sub, varnames = c("Var1", "Var2"), value.name = "CramersV")

  ggplot(df, aes(Var1, Var2, fill = CramersV)) +
    geom_tile(color = "white") +
    scale_fill_gradient2(
      low = "#17497b", mid = "#F7F7F7", high = "#b21880",
      midpoint = 0.30, limits = c(0, 1),
      name = "Cramér's V", na.value = "grey90"
    ) +
    labs(title = title, x = NULL, y = NULL) +
    theme_minimal(base_size = 14) +
    theme(
      axis.text.x = element_text(angle = 45, hjust = 1),
      panel.grid = element_blank()
    )
}

### 4. Create each heatmap
heatmap_body <- plot_heatmap(V_body, "Heatmap A - Body Part Application Correlations")
heatmap_where <- plot_heatmap(V_where, "Heatmap B - Dispenser Location Correlations")
heatmap_reason <- plot_heatmap(V_reason, "Heatmap C - Reasons Against Applying Sunscreen")

### 5. Show the plots
heatmap_body
heatmap_where
heatmap_reason
```

```
# Example: Check co-occurrence directly
table(dat_labeled$`Feels unpleasant`, dat_labeled$`Worried about skin reaction`)
```

```
# Or compute Jaccard similarity: overlap / union
```

```
overlap <- sum(dat_labeled$`Feels unpleasant` == 1 & dat_labeled$`Worried about skin reaction` == 1, na.rm = TRUE)
union <- sum(dat_labeled$`Feels unpleasant` == 1 | dat_labeled$`Worried about skin reaction` == 1, na.rm = TRUE)
jaccard <- overlap / union
jaccard
```

```

# Create a clean dataset with dummy columns added
dat_incentive <- dat_labeled %>%
  mutate(
    inc_sunburn = if_else(!is.na(incentive_lab) & incentive_lab == "Prevent sunburn", 1L, 0L),
    inc_cancer  = if_else(!is.na(incentive_lab) & incentive_lab == "Prevent skin cancer", 1L, 0L),
    inc_wrinkles = if_else(!is.na(incentive_lab) & incentive_lab == "Prevent wrinkles", 1L, 0L),
    inc_other   = if_else(!is.na(incentive_lab) & incentive_lab == "Other", 1L, 0L)
  )

body_cols <- intersect(label_body_parts(), names(dat_incentive))

# Cramér's V helper for binary vs binary
cramers_v_bin <- function(x, y){
  ok <- complete.cases(x, y)
  if (!any(ok)) return(NA_real_)
  tbl <- table(x[ok], y[ok])
  if (min(dim(tbl)) < 2) return(NA_real_)
  vcd::assocstats(tbl)$cramer
}

# Incentive columns
inc_cols <- c("inc_sunburn", "inc_cancer", "inc_wrinkles")

# Create empty matrix
V_inc_body <- matrix(NA_real_, nrow = length(inc_cols), ncol = length(body_cols),
  dimnames = list(gsub("^inc_", "", inc_cols), body_cols))

# Fill the matrix
for (i in seq_along(inc_cols)) {
  for (j in seq_along(body_cols)) {
    V_inc_body[i, j] <- cramers_v_bin(dat_incentive[[inc_cols[i]]], dat_incentive[[body_cols[j]]])
  }
}

V_inc_body_df <- melt(V_inc_body, varnames = c("Incentive", "Body"), value.name = "V")

ggplot(V_inc_body_df, aes(Body, Incentive, fill = V)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(low = "#17497b", mid = "#F7F7F7", high = "#b21880",
    midpoint = 0.15, limits = c(0,1), na.value = "grey90",
    name = "Cramér's V") +
  theme_minimal(base_size = 13) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
    panel.grid = element_blank()) +
  labs(title = "Association: Incentives × Body-part Selection",
    x = NULL, y = NULL)
V_inc_body_df %>%
  arrange(desc(V)) %>%
  head(10)

# Helper function: proportion selecting body part by incentive
prop_body_by_incentive <- function(df, inc_col, body_col) {
  df %>%

```

```

mutate(inc = .data[[inc_col]] == 1,
       body = .data[[body_col]] == 1) %>%
filter(!is.na(inc), !is.na(body)) %>%
summarise(
  incentive = inc_col,
  body_part = body_col,
  n_inc = sum(inc),
  n_noinc = sum(!inc),
  prop_body_if_inc = mean(body[inc]),
  prop_body_if_noinc = mean(body[!inc]),
  diff = prop_body_if_inc - prop_body_if_noinc,
  .groups = "drop"
)
}

# Get proportions for all incentive-body pairs
prop_results <- expand_grid(
  inc = inc_cols,
  body = body_cols,
  stringsAsFactors = FALSE
) %>%
rowwise() %>%
do(prop_body_by_incentive(dat_incentive, .$inc, .$body)) %>%
ungroup() %>%
arrange(desc(diff))

# Show top 15 pairs by largest difference in proportions
prop_results %>%
select(incentive, body_part, prop_body_if_inc, prop_body_if_noinc, diff) %>%
head(15)

```

```

library(dplyr)
library(tidyr)

# Incentive columns
inc_cols <- c("inc_sunburn", "inc_cancer", "inc_wrinkles")

# Body part columns
body_cols <- intersect(label_body_parts(), names(dat_incentive))

# Build proportions table
raw_props <- expand_grid(incentive = inc_cols, body_part = body_cols) %>%
rowwise() %>%
mutate(
  # People who selected the incentive
  prop_body_if_inc = mean(dat_incentive[[body_part]][dat_incentive[[incentive]] == 1] == 1, na.rm = TRUE),
  # People who did NOT select the incentive
  prop_body_if_noinc = mean(dat_incentive[[body_part]][dat_incentive[[incentive]] == 0] == 1, na.rm = TRUE),
  # Difference between proportions
  diff = prop_body_if_inc - prop_body_if_noinc
) %>%
ungroup()

```

```

library(dplyr)
library(tidyr)
library(ggplot2)

## 0) Build incentive dummies (skip if you already have them)
dat_incentive <- dat_labeled %>%
  mutate(
    inc_sunburn = if_else(!is.na(incentive_lab) & incentive_lab == "Prevent sunburn", 1L, 0L),
    inc_cancer  = if_else(!is.na(incentive_lab) & incentive_lab == "Prevent skin cancer", 1L, 0L),
    inc_wrinkles = if_else(!is.na(incentive_lab) & incentive_lab == "Prevent wrinkles", 1L, 0L)
  )

body_cols <- intersect(label_body_parts(), names(dat_incentive))
inc_cols  <- c("inc_sunburn", "inc_cancer", "inc_wrinkles")

## 1) Raw proportions (guard against empty groups)
raw_props <- tidyr::expand_grid(incentive = inc_cols, body_part = body_cols) %>%
  rowwise() %>%
  mutate(
    n_inc      = sum(dat_incentive[[incentive]] == 1, na.rm = TRUE),
    n_noinc    = sum(dat_incentive[[incentive]] == 0, na.rm = TRUE),
    n_body_inc = sum(dat_incentive[[incentive]] == 1 & dat_incentive[[body_part]] == 1, na.rm = TRUE),
    n_body_noinc = sum(dat_incentive[[incentive]] == 0 & dat_incentive[[body_part]] == 1, na.rm = TRUE),
    prop_body_if_inc = ifelse(n_inc > 0, n_body_inc / n_inc, NA_real_),
    prop_body_if_noinc = ifelse(n_noinc > 0, n_body_noinc / n_noinc, NA_real_),
    diff = prop_body_if_inc - prop_body_if_noinc
  ) %>%
  ungroup()

## 2) Keep only valid rows, then take up to 5 strongest
top5 <- raw_props %>%
  filter(!is.na(prop_body_if_inc), !is.na(prop_body_if_noinc)) %>%
  arrange(desc(diff)) %>%
  slice_head(n = 5)

## Optional: quick check
if (nrow(top5) == 0) {
  message("No valid rows to plot. Here are the top candidates before filtering:")
  print(raw_props %>% arrange(desc(diff)) %>% slice_head(n = 10))
}

## 3) Plot if there's anything to facet
if (nrow(top5) > 0) {
  plot_df <- top5 %>%
    select(incentive, body_part, prop_body_if_inc, prop_body_if_noinc) %>%
    pivot_longer(c(prop_body_if_inc, prop_body_if_noinc),
                 names_to = "group", values_to = "proportion") %>%
    mutate(
      group = ifelse(group == "prop_body_if_inc", "Selected Incentive", "Did Not Select"),
      incentive = gsub("^inc_", "", incentive)
    )

  ggplot(plot_df, aes(x = body_part, y = proportion, fill = group)) +

```

```

geom_bar(stat = "identity", position = position_dodge(0.8), width = 0.7) +
facet_wrap(~ incentive, scales = "free_y") +
scale_fill_manual(values = c("#17497b", "#b21880")) +
labs(
  title = "Top Incentive × Body-Part Differences (Proportions)",
  subtitle = "Each panel compares body-part selection among those who DID vs DID NOT choose the incentive",
  x = "Body Part", y = "Proportion", fill = "Group"
) +
theme_minimal(base_size = 14) +
theme(axis.text.x = element_text(angle = 45, hjust = 1),
      panel.grid.major.x = element_blank(),
      strip.text = element_text(face = "bold"))
}

```

```
glimpse(dat_labeled)
```

```
colnames(dat_labeled)
```

```

library(dplyr)
library(janitor)

# variables to one-hot (single-choice only)
fac_vars <- c(
  "school_lab", "gender_lab",
  "seen_dispenser_lab",
  "would_you_use_lab", "aid_in_prevention_lab",
  "skin_type_lab", "apply_freq_lab",
  "price_lab", "incentive_lab",
  "knowledge_lab"
)

# levels to skip globally (optional)
drop_levels <- c("Other", "Other Incentive", "Other Reason", "Other Location")

# Build 0/1 columns for ONE factor without dropping rows.
# NA in the source stays NA in all dummies for that variable.
make_dummies_one <- function(df, var, drop_levels = character()) {
  stopifnot(var %in% names(df))
  x <- df[[var]]
  if (!is.factor(x)) x <- factor(x)
  keep_lvls <- setdiff(levels(x), drop_levels)
  out <- setNames(
    as.data.frame(matrix(NA_integer_, nrow(df), length(keep_lvls))),
    paste0(var, "__", make_clean_names(keep_lvls))
  )
  for (k in seq_along(keep_lvls)) {
    lvl <- keep_lvls[k]
    out[[k]] <- ifelse(is.na(x), NA_integer_, as.integer(x == lvl))
  }
  out
}

# Build all blocks (one per variable)

```

```

dummy_blocks <- lapply(fac_vars, make_dummies_one, df = dat_labeled, drop_levels = drop_levels)

# DIAGNOSTIC: every block should have nrow == nrow(dat_labeled)
sizes <- sapply(dummy_blocks, nrow)
print(sizes) # should all be 1008
stopifnot(all(sizes == nrow(dat_labeled)))

# Bind them safely
dummies_all <- bind_cols(dummy_blocks) %>% clean_names()

# Add to your data
dat_labeled <- bind_cols(dat_labeled, dummies_all)

cat("Added", ncol(dummies_all), "dummy columns. Rows:", nrow(dat_labeled), "\n")

```

```

glimpse(dat_labeled)

```

```

colnames(dat_labeled)

```

```

library(dplyr)

```

```

# Automatically detect real binary columns: only 0, 1, and optional NA
binary_cols <- dat_labeled %>%
  select(where(~ all(na.omit(.) %in% c(0, 1)))) %>%
  select(-key_information_the_inform, -redcap_survey_identifider, -`Other Reason`, -`Other Location`) #

# Check the resulting binary variables
str(binary_cols)

```

```

library(vcd)
library(reshape2)
library(dplyr)

```

```

# Step 1: Select clean binary-only columns
binary_cols <- dat_labeled %>%
  select(where(~ all(na.omit(.) %in% c(0, 1)))) %>%
  select(
    -redcap_survey_identifider,
    -key_information_the_inform,
    -`Other Reason`,
    -`Other Location`
  )

# Step 2: Robust Cramér's V function
cramers_v_bin <- function(x, y){
  ok <- complete.cases(x, y)
  if (!any(ok)) return(NA_real_)
  tbl <- table(x[ok], y[ok])
  if (min(dim(tbl)) < 2) return(NA_real_)
  vcd::assocstats(tbl)$cramer
}

```

```

# Step 3: Create empty matrix
p <- ncol(binary_cols)
V_matrix <- matrix(NA_real_, nrow = p, ncol = p,
                   dimnames = list(names(binary_cols), names(binary_cols)))

# Step 4: Fill matrix
for (i in seq_len(p)) {
  for (j in seq_len(p)) {
    V_matrix[i, j] <- if (i == j) 1 else cramers_v_bin(binary_cols[[i]], binary_cols[[j]])
  }
}

# Step 5: Melt into long format
V_long <- melt(V_matrix, varnames = c("Var1", "Var2"), value.name = "CramersV")

# FIX: Convert to characters to avoid factor comparison issues
V_long <- V_long %>%
  mutate(
    Var1 = as.character(Var1),
    Var2 = as.character(Var2)
  ) %>%
  filter(Var1 < Var2) %>%      # Remove duplicates & self-pairs
  arrange(desc(CramersV))

# Step 6: View top associations

options(pager = "cat")

print(V_long, n = Inf)

```

```

library(dplyr)
library(vcd)      # assocstats()
library(purrr)    # map_dfr
library(tidyr)

# ---- 0) Build the dummy set you want to analyze -----
dummy_cols <- dat_labeled %>%
  select(
    # multi-select (drop Other*)
    `Exits/Entrances`:`Other Location`,
    Face:Feet,
    `Feels unpleasant`:`Other Reason`,
    # seasonal
    winter_1:fall_4,
    # your one-hot dummies
    starts_with("school_lab_"),
    starts_with("gender_lab_"),
    starts_with("seen_dispenser_lab_"),
    starts_with("would_you_use_lab_"),
    starts_with("aid_in_prevention_lab_"),
    starts_with("skin_type_lab_"),
    starts_with("apply_freq_lab_"),
    starts_with("price_lab_"),

```

```

    starts_with("incentive_lab_"),
    starts_with("knowledge_lab_")
  ) %>%
  select(-`Other Location`, -`Other Reason`) %>% # drop the "Other" multiselects
  # ensure 0/1/NA integers (this is where across() must be in mutate)
  mutate(across(everything(), ~ as.integer(.x)))

# ---- 1) Parent mapping so we can drop within-parent pairs -----
parent_of <- function(nm) {
  case_when(
    nm %in% c("Exits/Entrances", "Bathrooms", "Bus stops", "Parking lots",
              "Classrooms", "Student health", "Student lounge", "Athletic fields") ~ "where",
    nm %in% c("Face", "Head", "Neck", "Lips", "Ears", "Chest", "Abdomen",
              "Upper back", "Lower back", "Arms", "Hands", "Legs", "Feet") ~ "body",
    nm %in% c("Feels unpleasant", "Worried about skin reaction", "Cost",
              "Forget", "Feels unnecessary") ~ "reasons",
    grepl("^winter_|^spring_|^summer_|^fall_", nm) ~ "season",
    grepl("^school_lab_", nm) ~ "school",
    grepl("^gender_lab_", nm) ~ "gender",
    grepl("^seen_dispenser_lab_", nm) ~ "seen",
    grepl("^would_you_use_lab_", nm) ~ "would_use",
    grepl("^aid_in_prevention_lab_", nm) ~ "aid_prev",
    grepl("^skin_type_lab_", nm) ~ "skin_type",
    grepl("^apply_freq_lab_", nm) ~ "apply_freq",
    grepl("^price_lab_", nm) ~ "price",
    grepl("^incentive_lab_", nm) ~ "incentive",
    grepl("^knowledge_lab_", nm) ~ "knowledge",
    TRUE ~ "other"
  )
}

# ---- 2) Cramér's V for 2x2 + chi-square p-value -----
bin_cramer <- function(x, y) {
  ok <- stats::complete.cases(x, y)
  if (!any(ok)) return(c(V = NA_real_, p = NA_real_, n = 0))
  xt <- table(x[ok], y[ok])
  if (any(dim(xt) < 2)) return(c(V = NA_real_, p = NA_real_, n = sum(ok)))
  a <- suppressWarnings(vcd::assocstats(xt))
  ct <- suppressWarnings(chisq.test(xt, correct = FALSE))
  c(V = unname(a$cramer), p = unname(ct$p.value), n = sum(xt))
}

# ---- 3) All cross-parent pairs, V + p + FDR -----
nm <- names(dummy_cols)
idx <- combn(seq_along(nm), 2)

res <- map_dfr(
  seq_len(ncol(idx)),
  function(k) {
    i <- idx[1, k]; j <- idx[2, k]
    vv <- bin_cramer(dummy_cols[[i]], dummy_cols[[j]])
    tibble(
      Var1 = nm[i], Var2 = nm[j],

```

```

    parent1 = parent_of(nm[i]), parent2 = parent_of(nm[j]),
    V = as.numeric(vv["V"]),
    p = as.numeric(vv["p"]),
    n = as.integer(vv["n"])
  )
}
) %>%
  filter(parent1 != parent2, !is.na(V)) %>%
  mutate(p_adj_fdr = p.adjust(p, method = "fdr")) %>%
  arrange(desc(V), p_adj_fdr)

# ---- 4) Print the top K without paging -----
options(pager = "cat")
top_k <- 4000 # change to any size you want
print(res %>% select(Var1, Var2, V, p, p_adj_fdr, n) %>% slice_head(n = top_k), n = top_k)

```

```

library(dplyr)
library(vcd)
library(tidyr)
library(purrr)

# ---- 1) Pick the variables to test (use their labeled forms!) ----
vars <- dat_labeled %>%
  select(
    school_lab,
    gender_lab,
    seen_dispenser_lab,
    would_you_use_lab,
    aid_in_prevention_lab,
    skin_type_lab,
    apply_freq_lab,
    price_lab,
    incentive_lab,
    knowledge_lab
  )

# ---- 2) Function for Cramér's V + p-value between two full variables ----
var_cramer <- function(x, y) {
  ok <- complete.cases(x, y)
  if (!any(ok)) return(c(V = NA_real_, p = NA_real_, n = 0))
  xt <- table(x[ok], y[ok])
  if (any(dim(xt) < 2)) return(c(V = NA_real_, p = NA_real_, n = sum(ok)))
  assoc <- suppressWarnings(vcd::assocstats(xt))
  test <- suppressWarnings(chisq.test(xt, correct = FALSE))
  c(V = unname(assoc$cramer), p = unname(test$p.value), n = sum(xt))
}

# ---- 3) Compute all pairwise variable-to-variable associations ----
nm <- names(vars)
idx <- combn(seq_along(nm), 2)

res_var <- map_dfr(
  seq_len(ncol(idx)),

```

```

function(k) {
  i <- idx[1, k]; j <- idx[2, k]
  vv <- var_cramer(vars[[i]], vars[[j]])
  tibble(
    Var1 = nm[i], Var2 = nm[j],
    V = as.numeric(vv["V"]),
    p = as.numeric(vv["p"]),
    n = as.integer(vv["n"])
  )
}
) %>%
  filter(!is.na(V)) %>%
  mutate(p_adj_fdr = p.adjust(p, method = "fdr")) %>%
  arrange(desc(V), p_adj_fdr)

# ---- 4) Output all results ----
options(pager = "cat")
print(res_var %>% select(Var1, Var2, V, p, p_adj_fdr, n), n = nrow(res_var))

```